Team 06:

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Github Repo URL: https://github.com/adi776borate/CS203-STT-Labs/tree/main/CS203_Lab_08

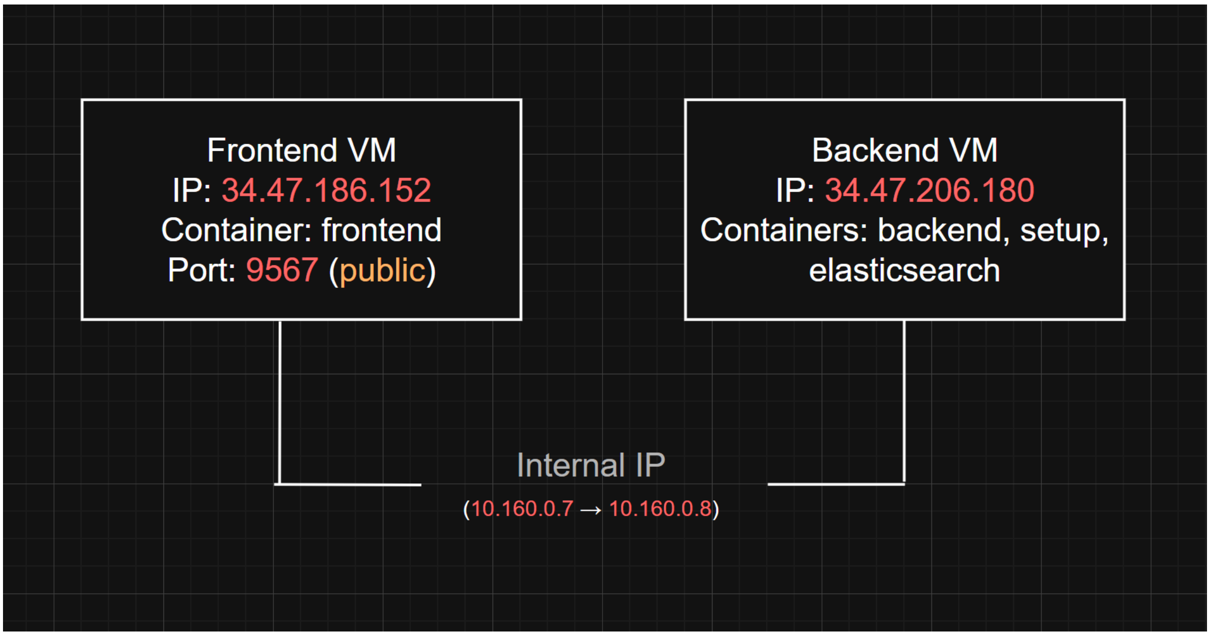
Docker Hub Repo URLs:

- fastapi: adi362/fastapi | Docker Hub
- elasticsearch: adi362/elasticsearch general | Docker Hub
- setup: adi362/setup | Docker Hub

Dockerfile URL: https://drive.google.com/drive/folders/18eJoLZKfgNLAzVrL1IYCt9qDmpgTlwg5?usp=sharing

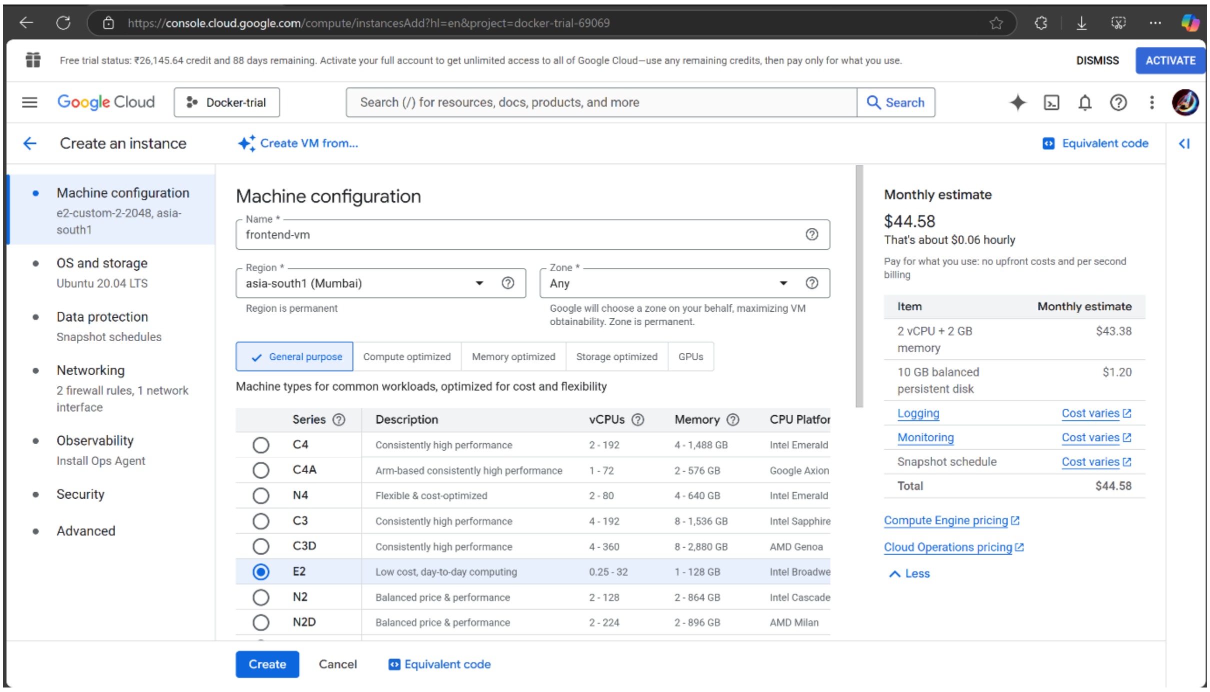
🔷 Overview of the Assignemnt:

This assignment involves deploying a multi-container Dockerized application using FastAPI and Elasticsearch across two separate virtual machines on GCP. The frontend must be public and isolated in a different VM, while the backend and Elasticsearch reside together on another VM.

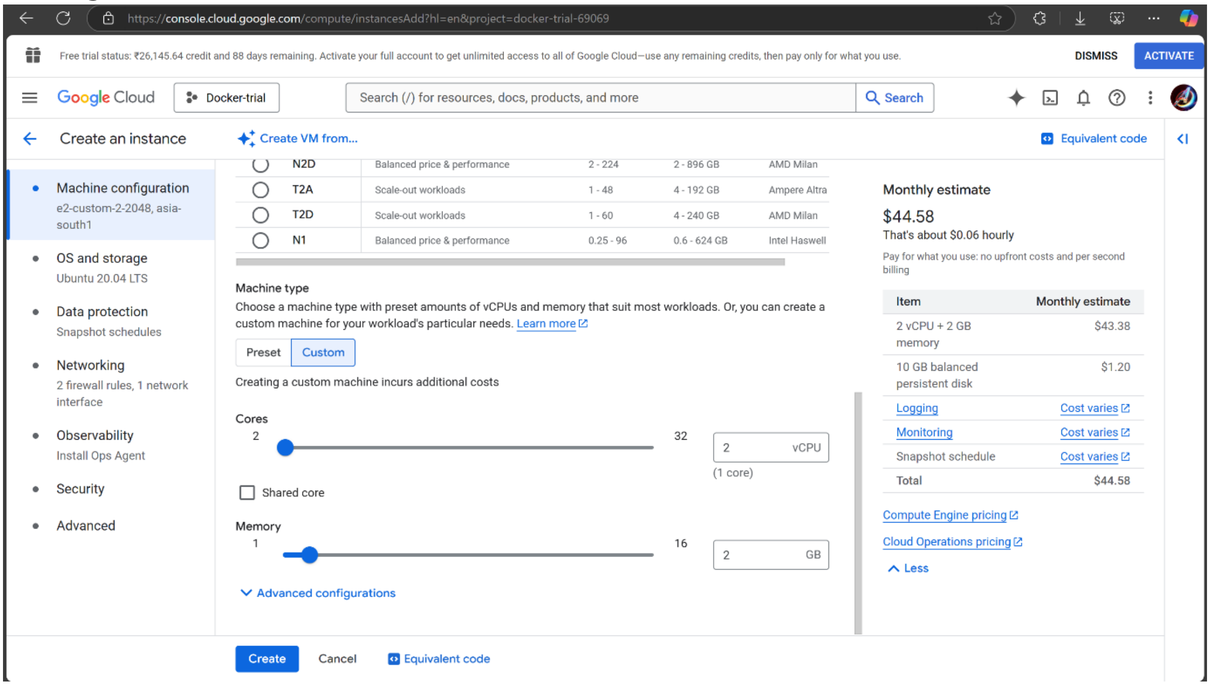


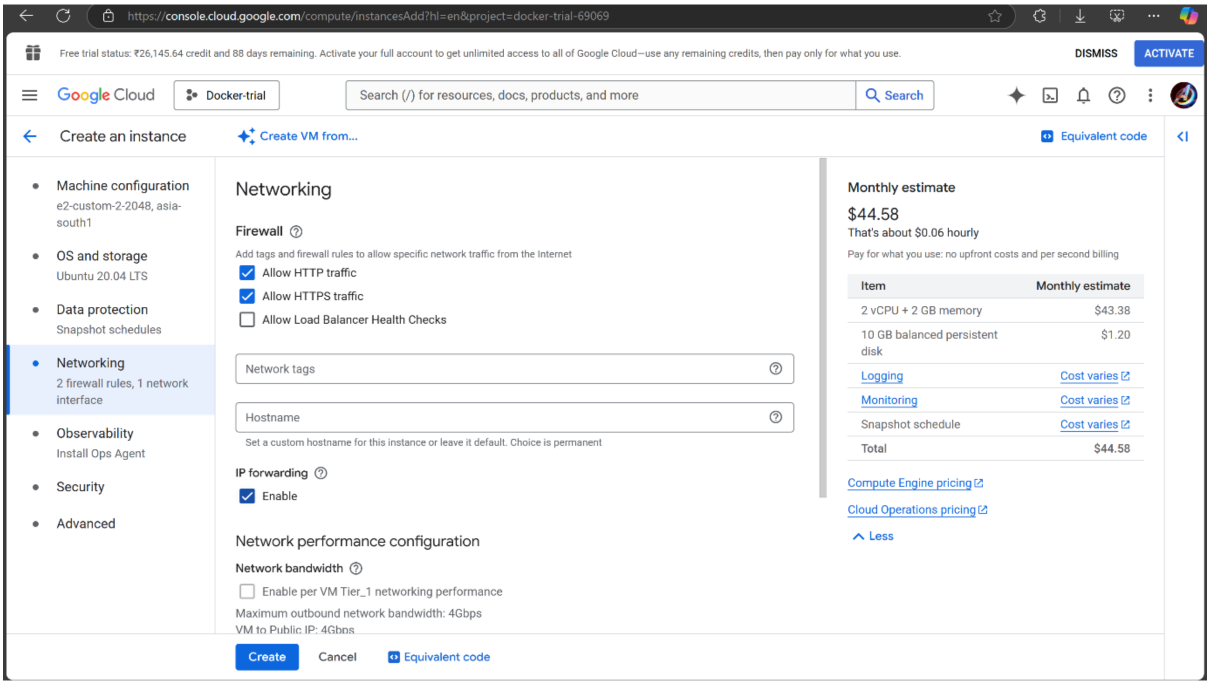
✅ VM Instance Creation

1. Creating a VM instance for the frontend container (frontend-vm) on Google Cloud Platform.

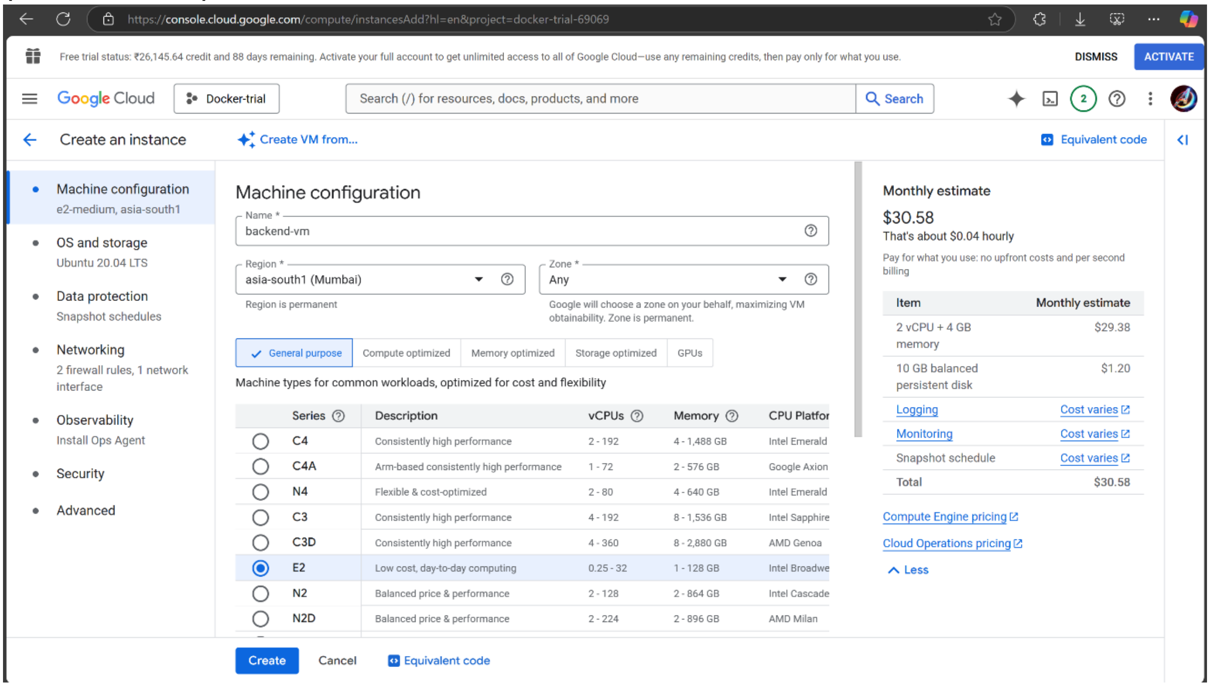


2. Configuring machine type (E2), boot disk (Ubuntu), and firewall settings during VM creation.

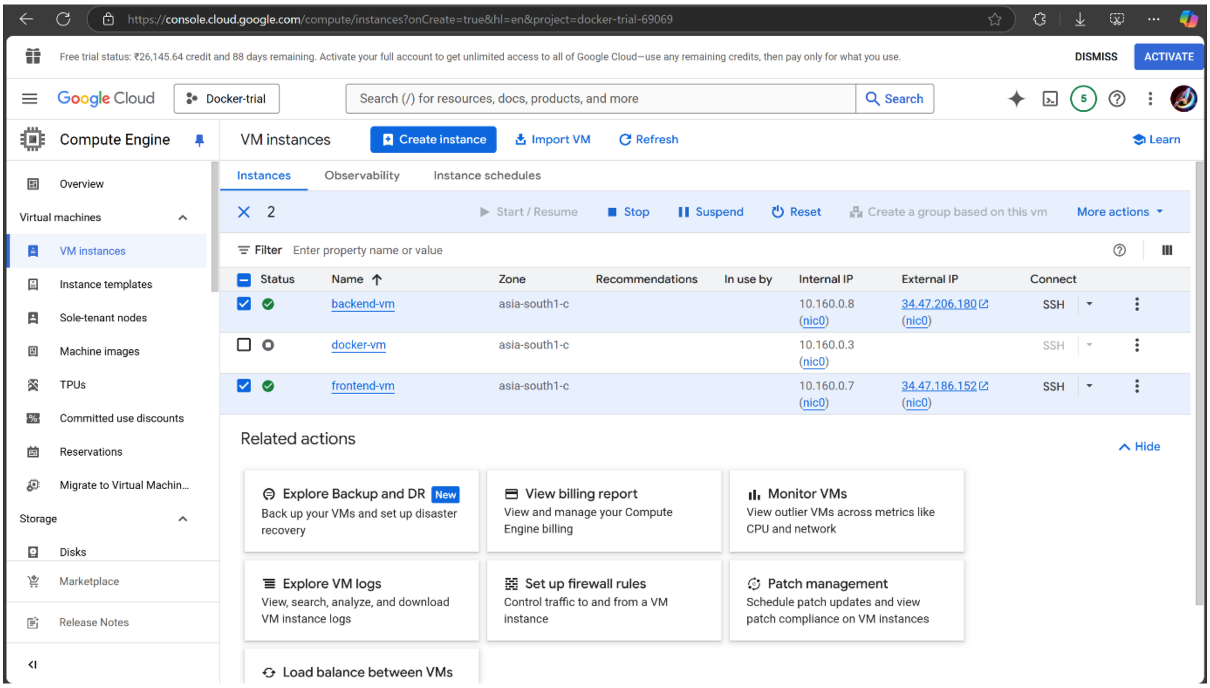




3. Creating a VM instance for the backend and Elasticsearch services (backend-vm) on GCP.

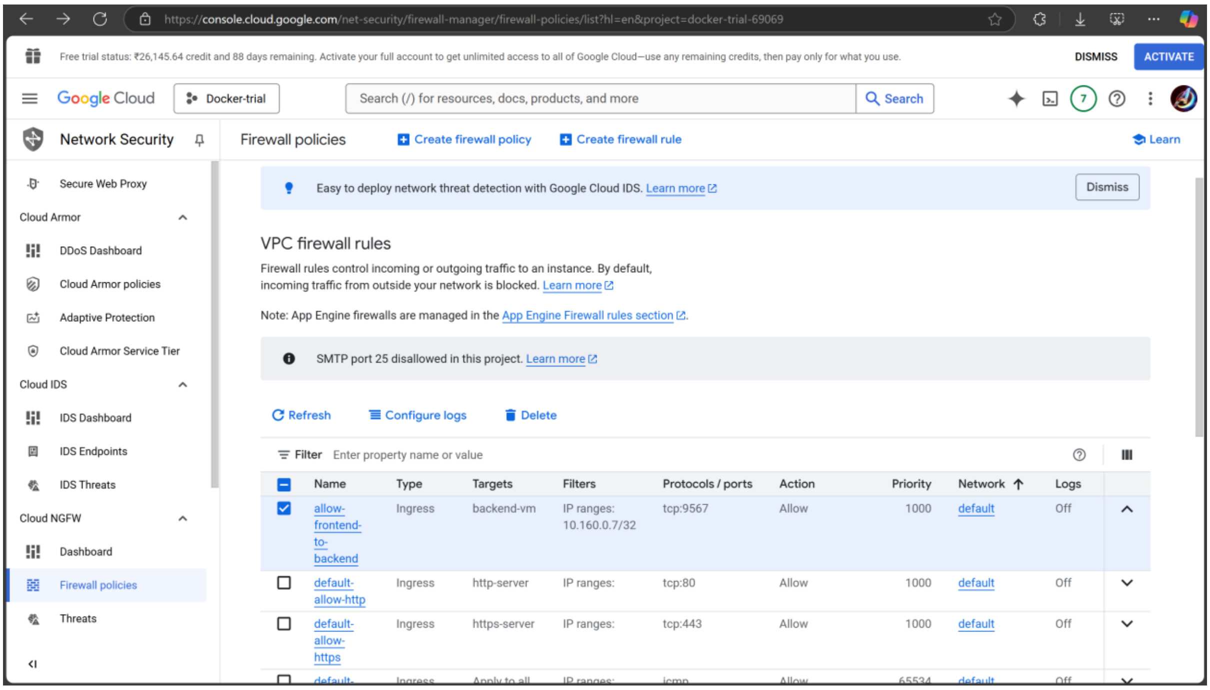


4. Two VM instances created

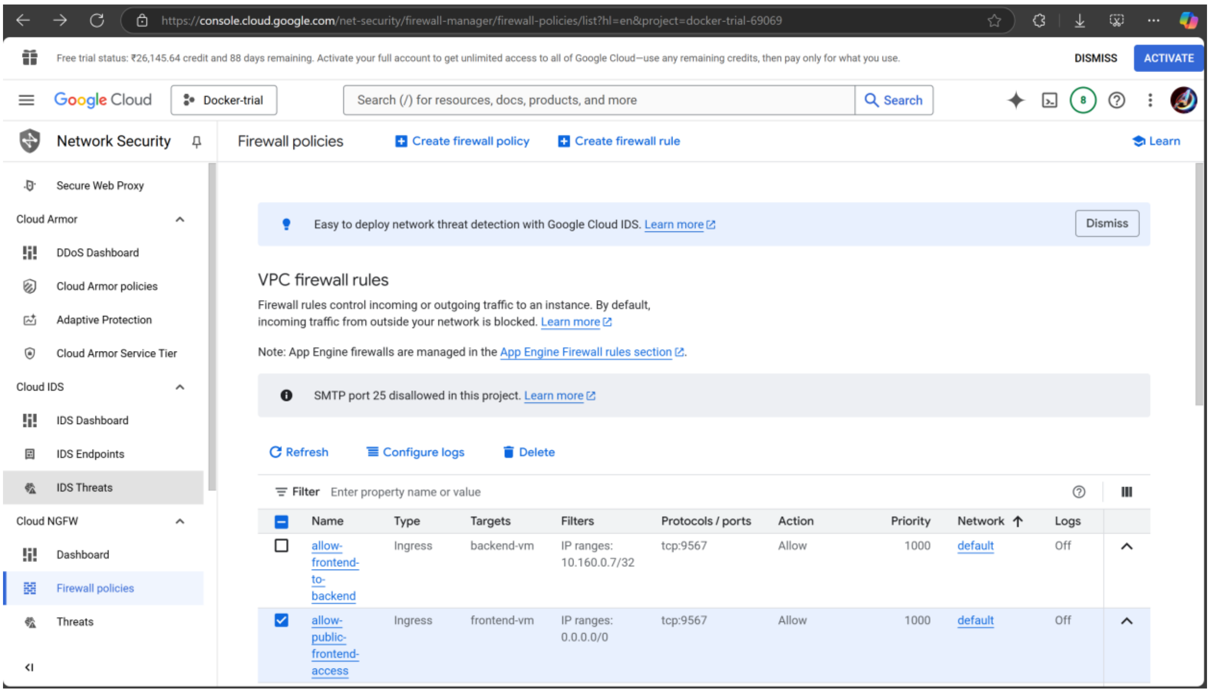


✅ Adding 2 Firewall Rules

1. Created a custom firewall rule to restrict backend access to only the frontend VM

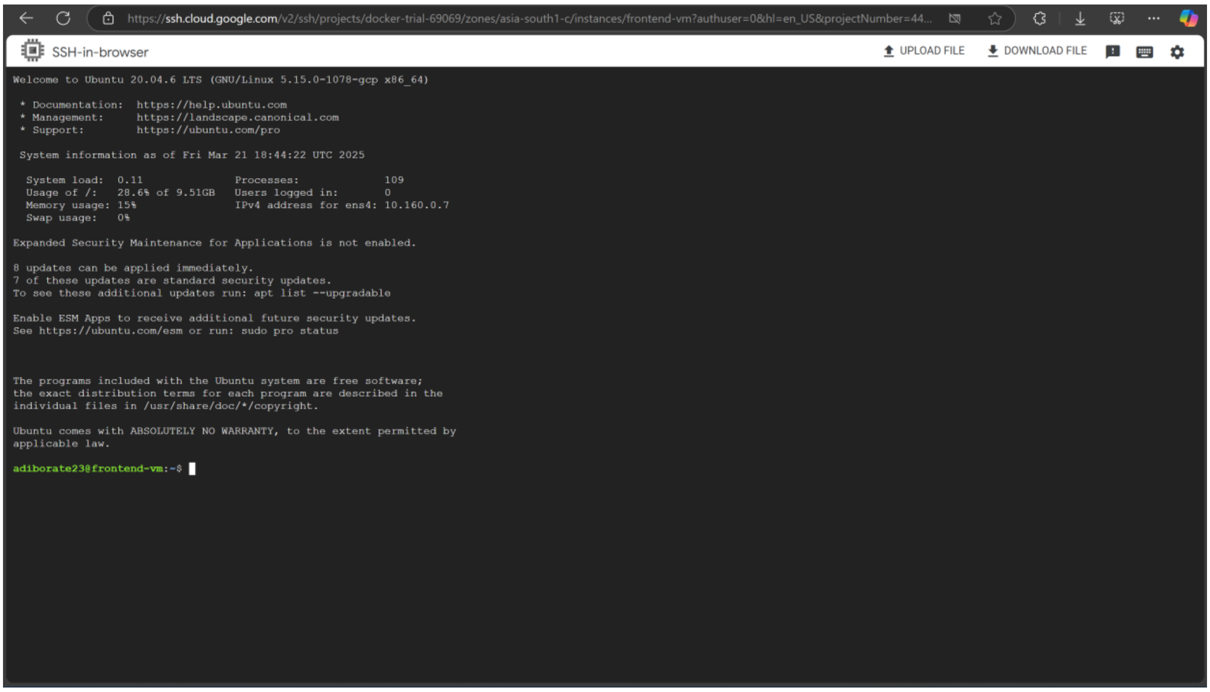


2. Created another custom firewall rule to allow public frontend access



✅ Running frontend-vm by connecting SSH

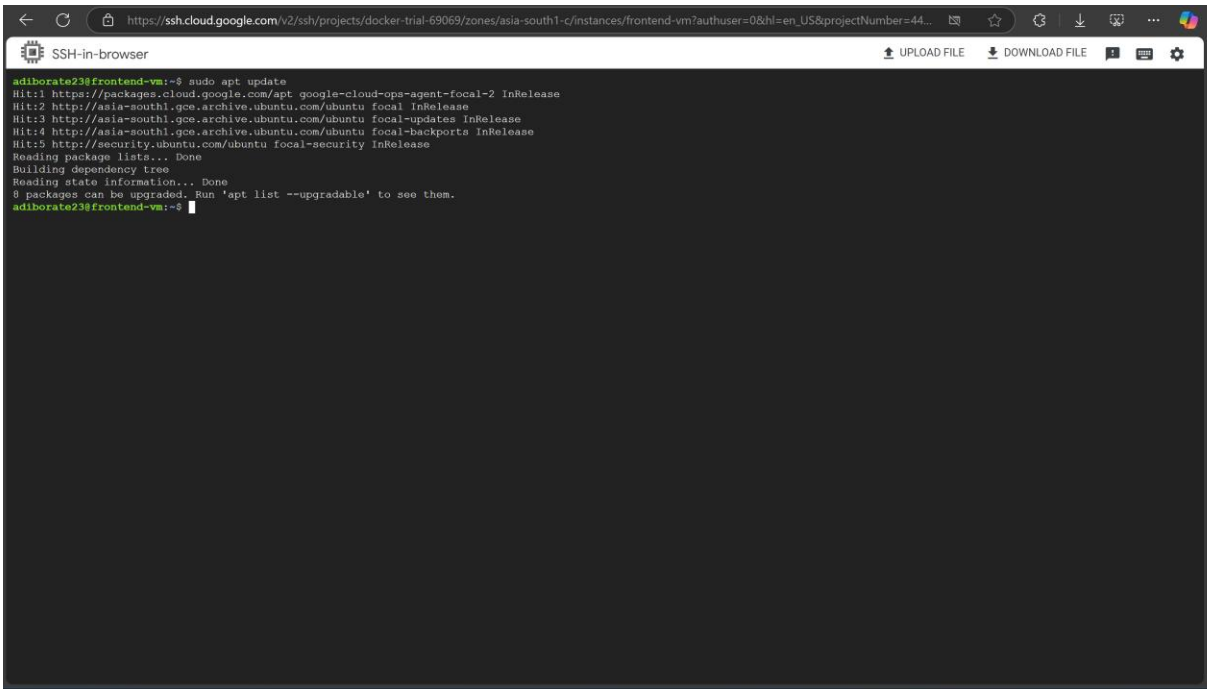
1. Initial stage:



2. Installing Docker:

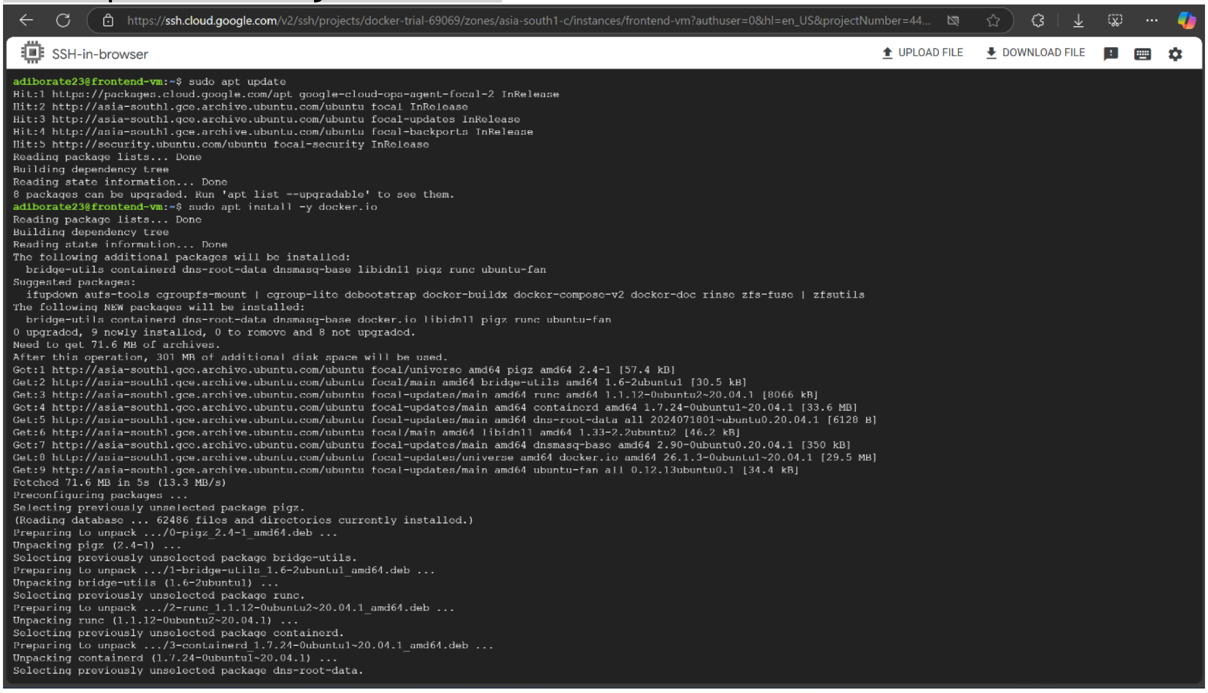
Update package list

```
sudo apt update
```



Install Docker

```
sudo apt install -y docker.io
```



Enable Docker service

```
sudo systemctl enable docker
```

Start Docker service

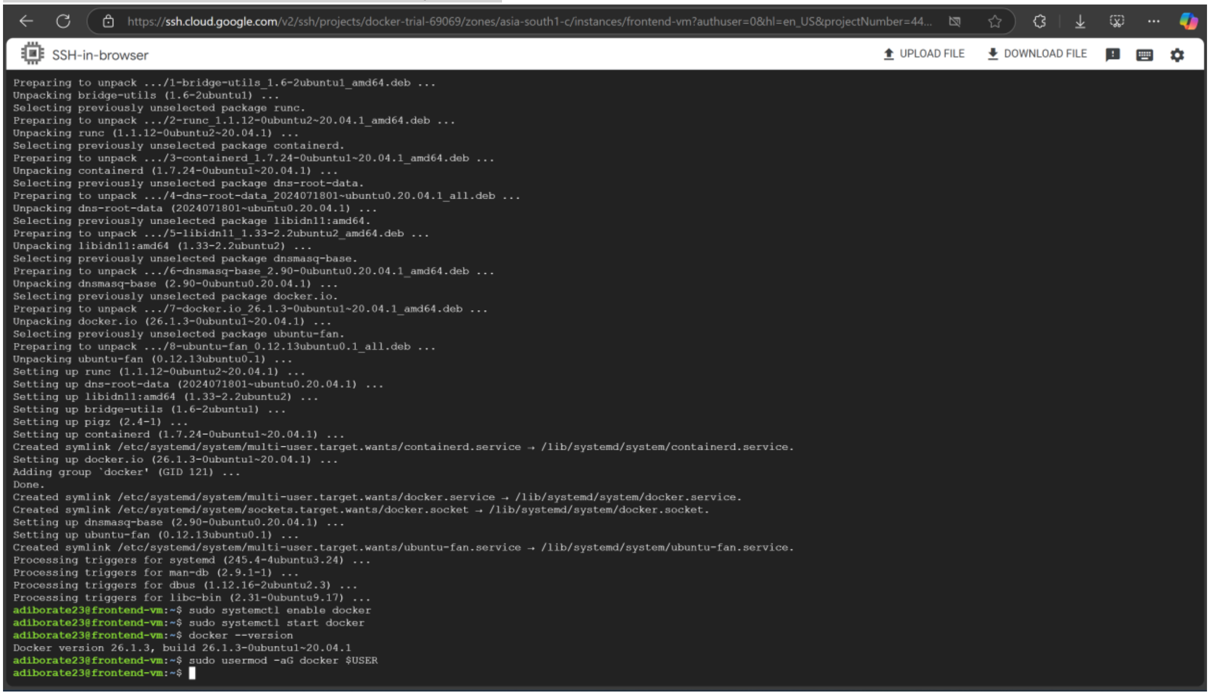
```
sudo systemctl start docker
```

Verify Docker installation

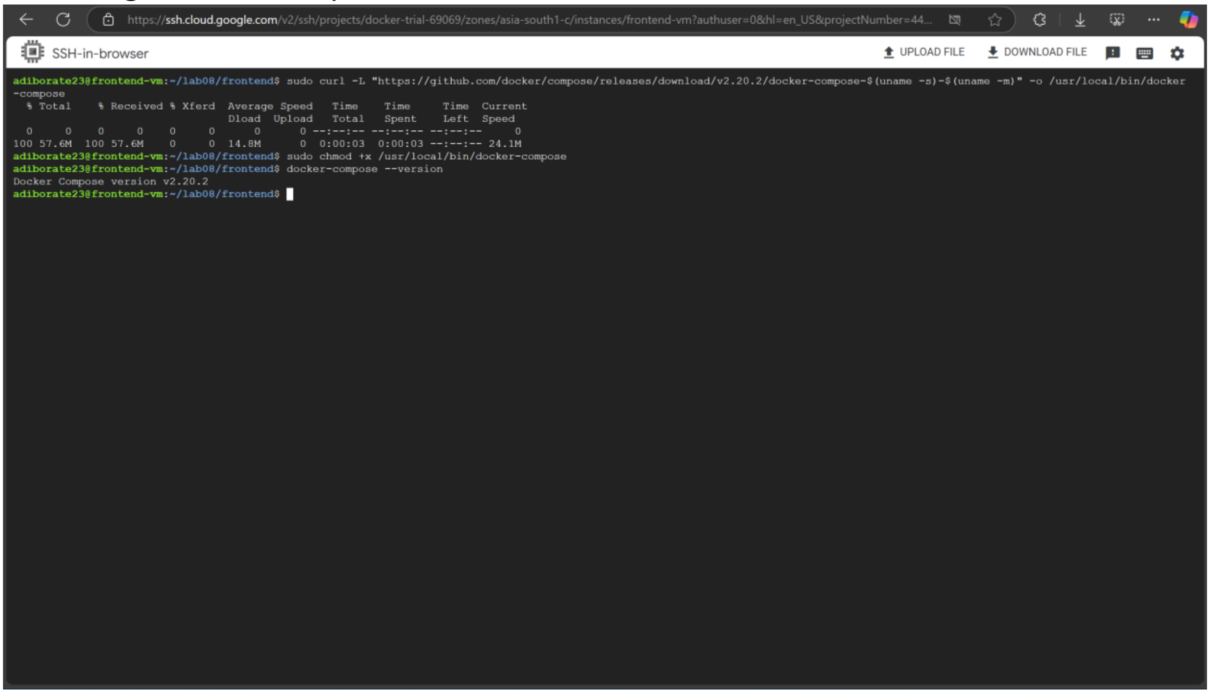
```
docker --version
```

#Add Your User to the Docker Group

```
sudo usermod -aG docker $USER
```

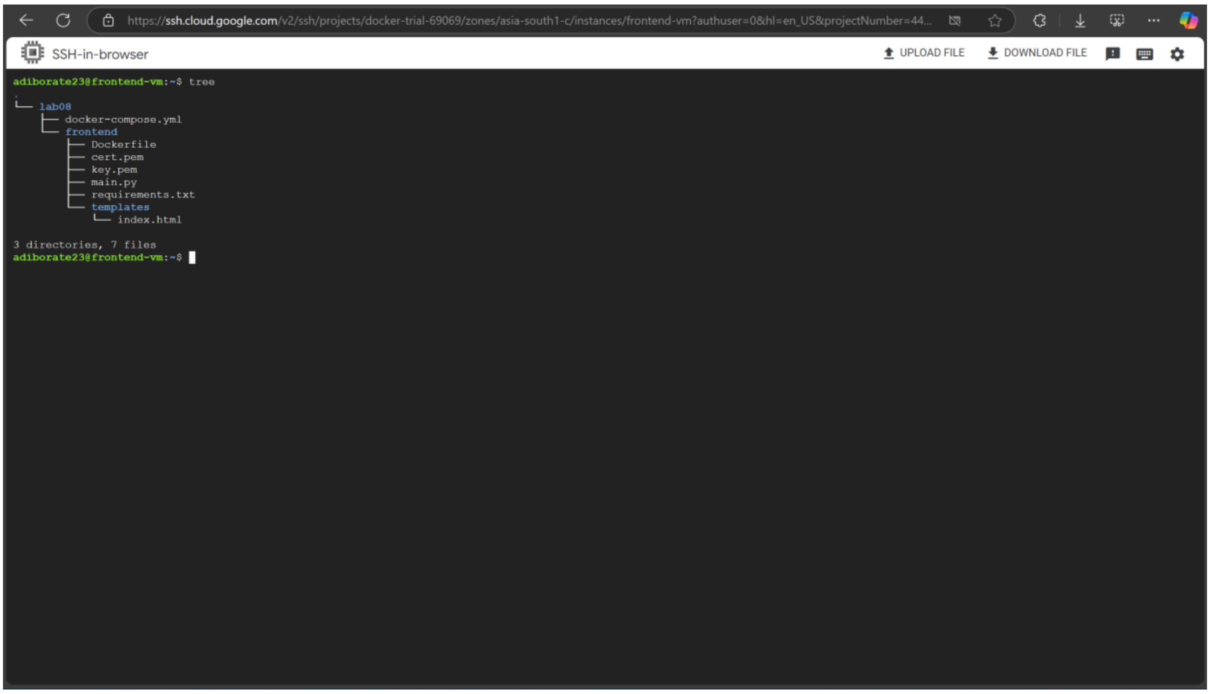


3. Installing docker-compose



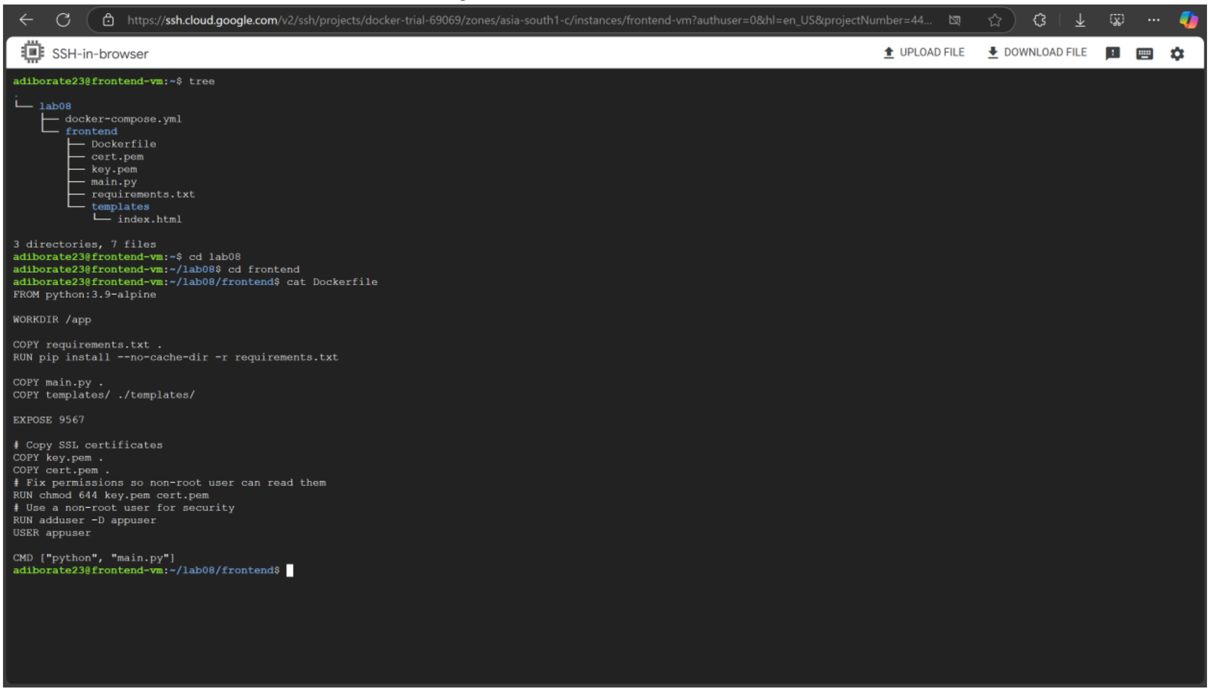
✅ Setting Up Frontend

1. After writing necessary frontend files, this is how the directory structure is:



FastAPI, by default, works with HTTP and requires configuration to use HTTPS. The cert.pem and key.pem are the certificate files for setting up HTTPS access, generated with this command:
`openssl req -x509 -newkey rsa:4096 -nodes -out cert.pem -keyout key.pem -days 365`

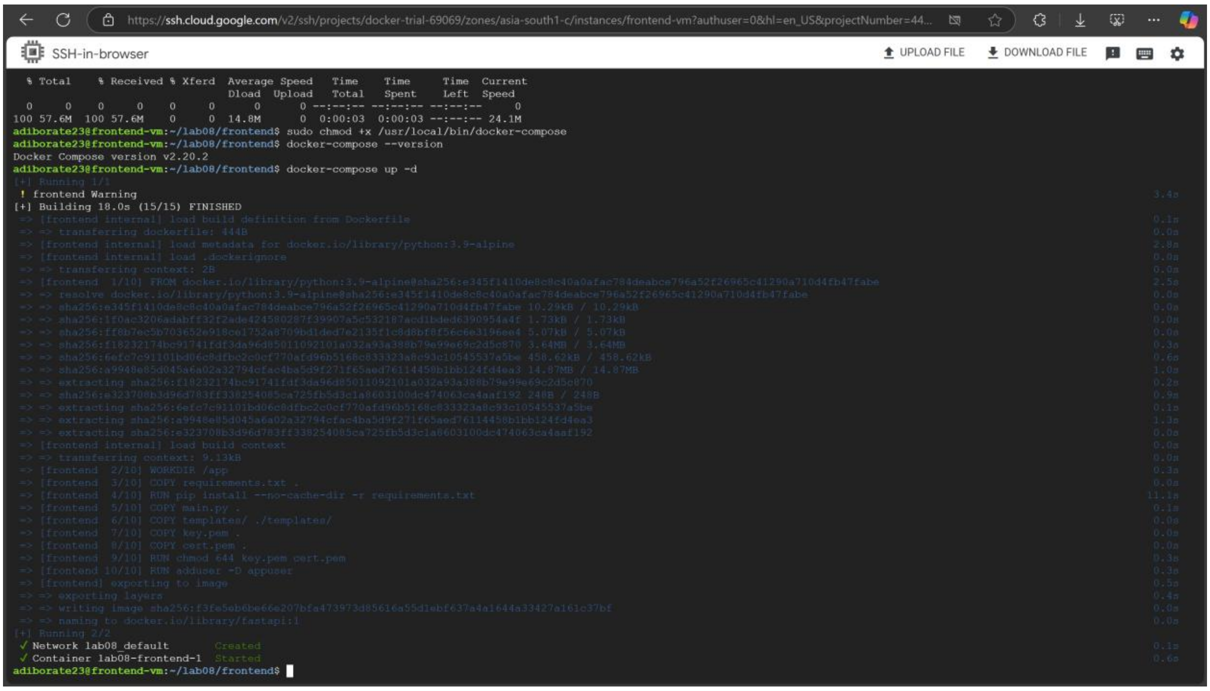
2. Dockerfile in frontend directory:



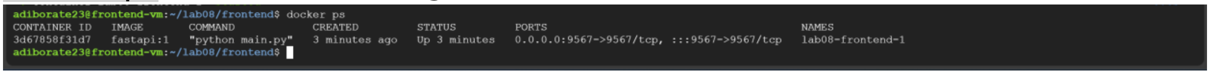
(To view entire codebase, please refer to the attached GitHub Repository URL at the top)

✅ Build and start the frontend container

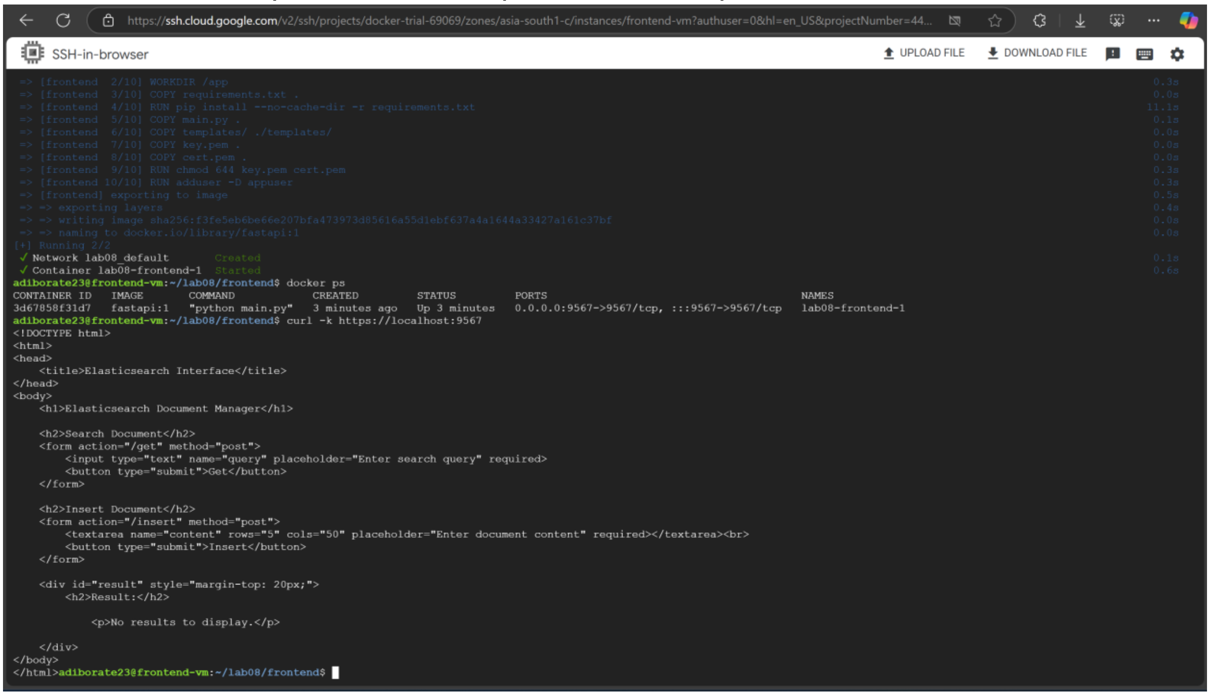
1. `docker-compose up -d` : builds (if needed) and starts the frontend container in detached mode, setting it up with the defined ports, volumes, and environment as specified in docker-compose.yml



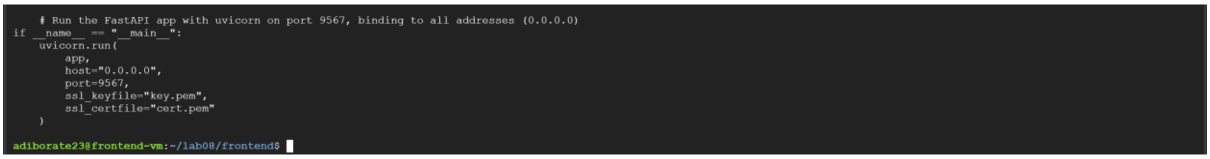
2. `docker ps` : shows all running containers



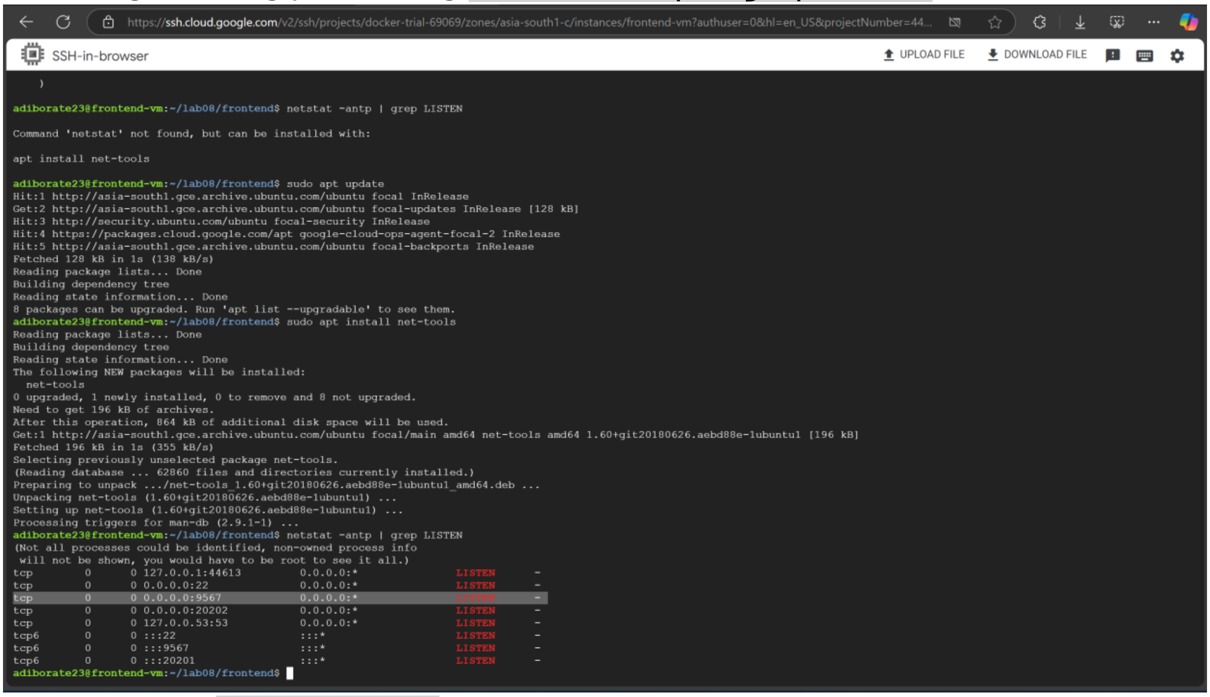
3. Test frontend endpoint with curl (internal check)



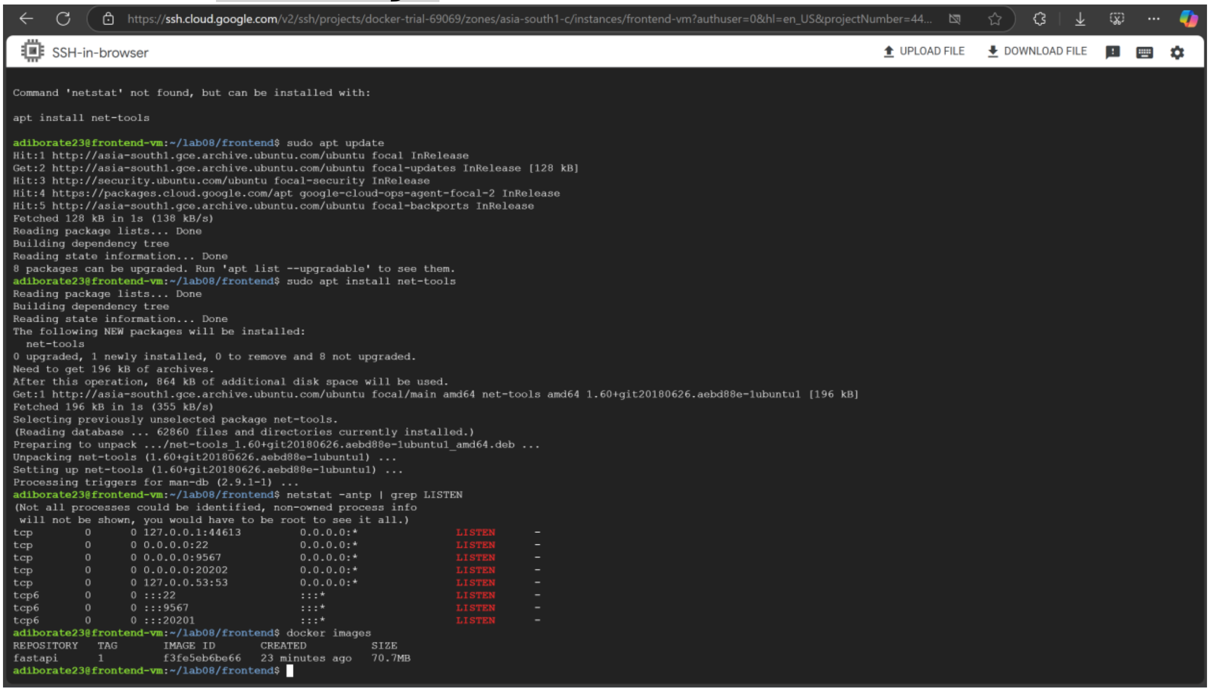
4. App is bound to `0.0.0.0` in code to allow external access using this part of code in main.py



5. Checking listening ports using `netstat -antp | grep LISTEN`



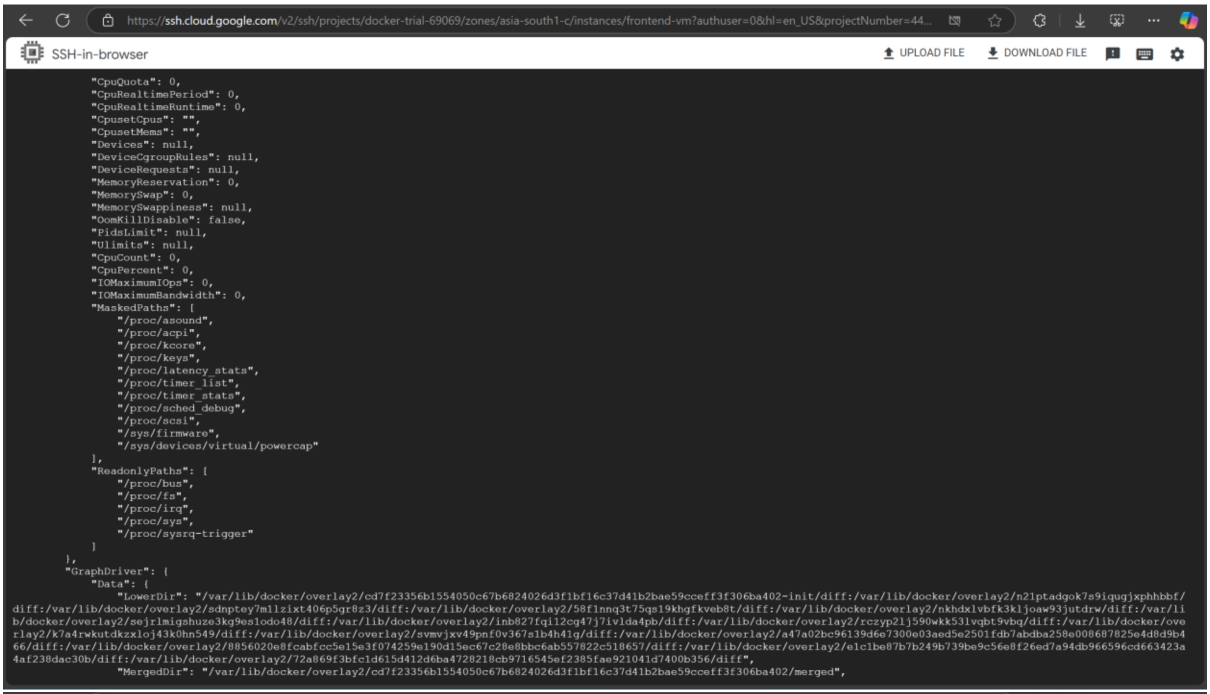
6. Screenshot of `docker images`

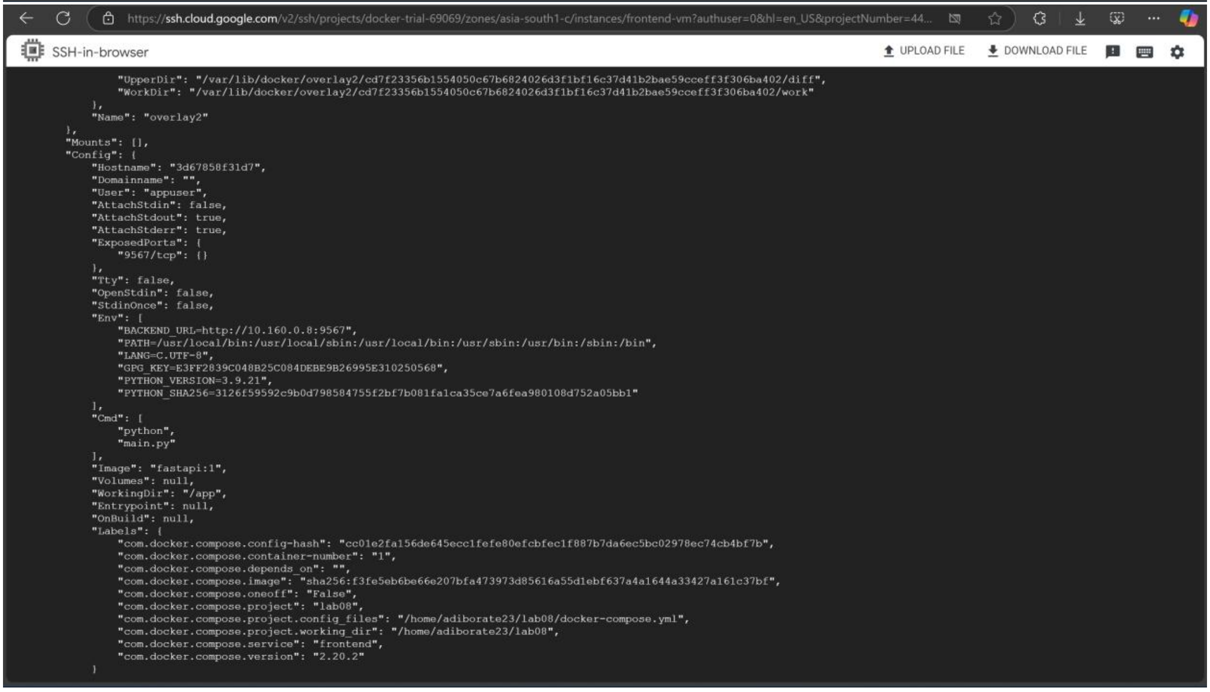


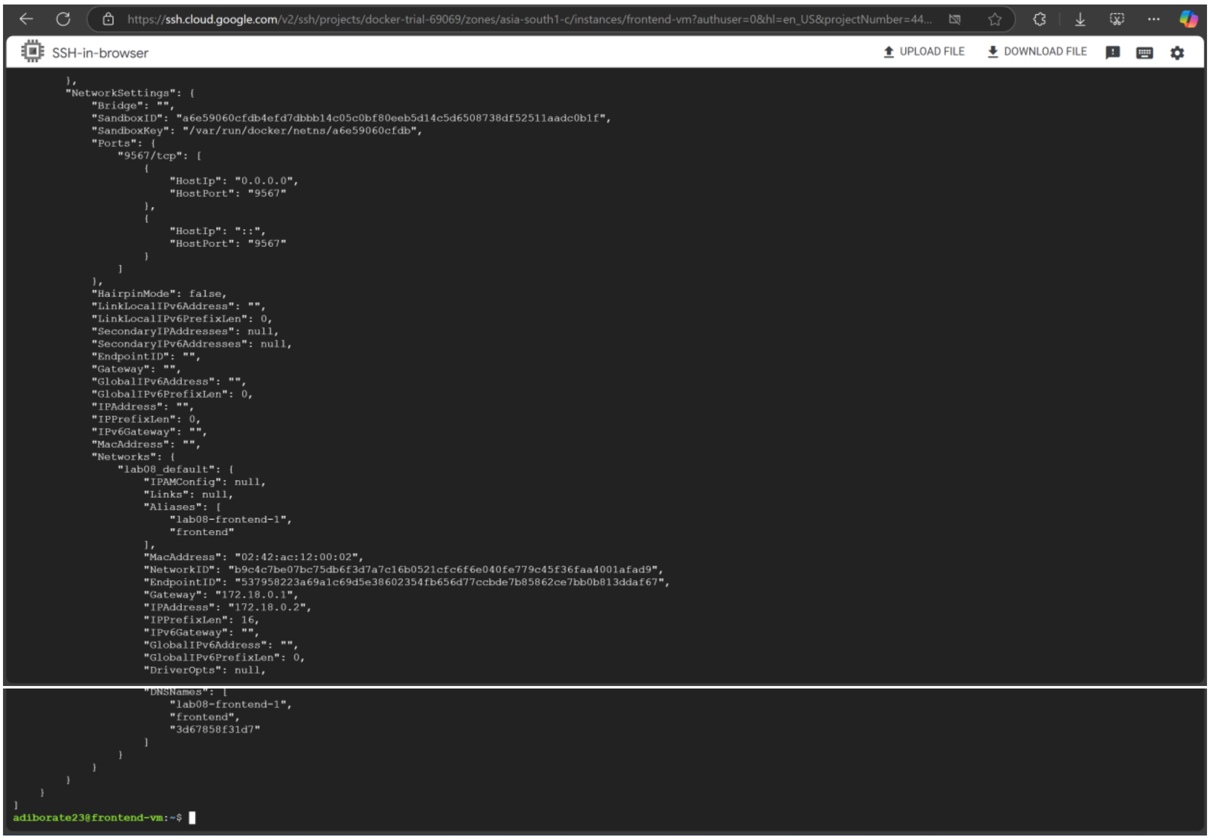
7. `docker inspect` on frontend container

```
adiborate23@frontend-vm:~$ docker inspect 3d67858f31d7
[
    {
        "Id": "3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869",
        "Created": "2025-03-22T05:20:46.677026827Z",
        "Path": "python",
        "Args": [
            "main.py"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 53904,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2025-03-22T05:20:47.219868357Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:f3fe5eb6be66e207bfa473973d85616a55d1ebf637a4a1644a33427a161c37bf",
        "ResolvConfPath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/hostname",
        "HostsPath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/hosts",
        "LogPath": "/var/lib/docker/containers/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869/3d67858f31d771b3a29c7a5f13cc43e16847873d96a6b12161abf00429d82869-jso
n.log",
        "Name": "/lab08-frontend-1",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "docker-default",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "lab08_default",
            "PortBindings": {
                "9567/tcp": [
                    {
                        "HostIp": "",
```

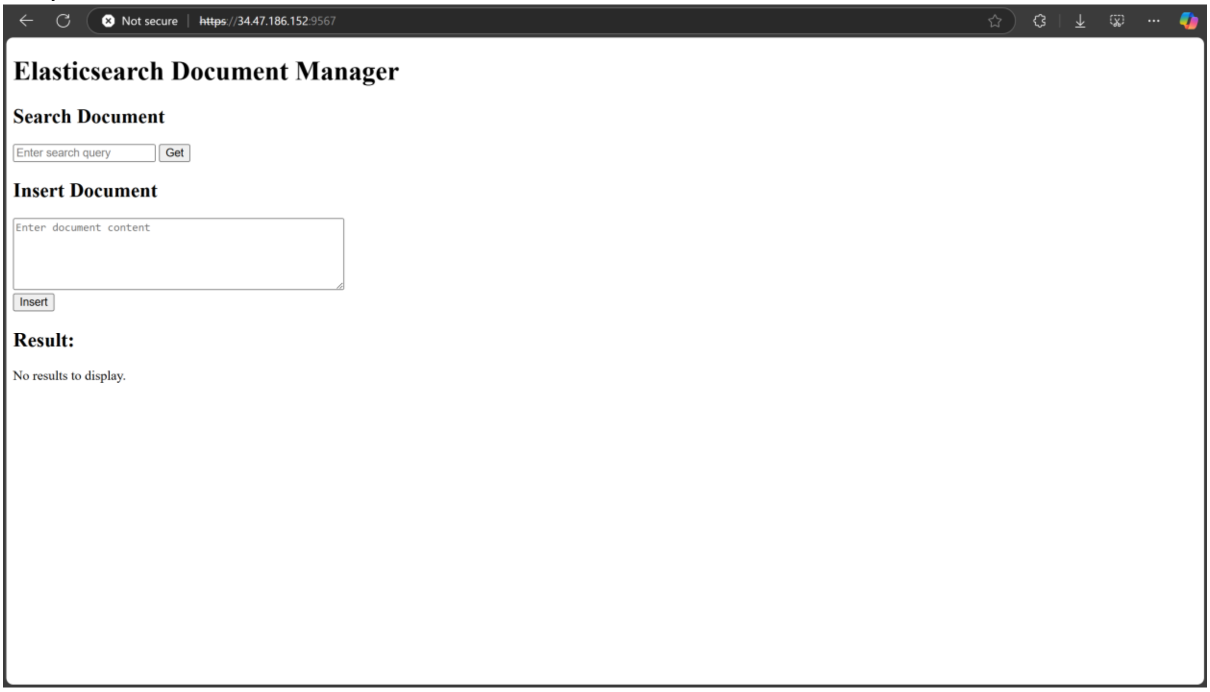
```
                        "HostPort": "9567"
                    }
                ]
            },
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
            "ConsoleSize": [
                0,
                0
            ],
            "CapAdd": null,
            "CapDrop": null,
            "CgroupnsMode": "host",
            "Dns": null,
            "DnsOptions": null,
            "DnsSearch": null,
            "ExtraHosts": [],
            "GroupAdd": null,
            "IpcMode": "private",
            "Cgroup": "",
            "Links": null,
            "OomScoreAdj": 0,
            "PidMode": "",
            "Privileged": false,
            "PublishAllPorts": false,
            "ReadonlyRootfs": false,
            "SecurityOpt": null,
            "UTSMode": "",
            "UsernsMode": "",
            "ShmSize": 67108864,
            "Runtime": "runc",
            "Isolation": "",
            "CpuShares": 0,
            "Memory": 0,
            "NanoCpus": 0,
            "CgroupParent": "",
            "BlkioWeight": 0,
            "BlkioWeightDevice": null,
            "BlkioDeviceReadBps": null,
            "BlkioDeviceWriteBps": null,
            "BlkioDeviceReadIOps": null,
            "BlkioDeviceWriteIOps": null,
            "CpuPeriod": 0,
```


https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/frontend-vm?authuser=0&hl=en_US&projectNumber=44...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
"CpuQuota": 0,
"CpuRealtimePeriod": 0,
"CpuRealtimeRuntime": 0,
"CpusetCpus": "",
"CpusetMems": "",
"Devices": null,
"DeviceCgroupRules": null,
"DeviceRequests": null,
"MemoryReservation": 0,
"MemorySwap": 0,
"MemorySwappiness": null,
"OomKillDisable": false,
"PidsLimit": null,
"Ulimits": null,
"CpuCount": 0,
"CpuPercent": 0,
"IOMaximumIOps": 0,
"IOMaximumBandwidth": 0,
"MaskedPaths": [
"/proc/asound",
"/proc/acpi",
"/proc/kcore",
"/proc/keys",
"/proc/latency_stats",
"/proc/timer_list",
"/proc/timer_stats",
"/proc/sched_debug",
"/proc/scsi",
"/sys/firmware",
"/sys/devices/virtual/powercap"
],
"ReadonlyPaths": [
"/proc/bus",
"/proc/fs",
"/proc/irq",
"/proc/sys",
"/proc/sysrq-trigger"
]
},
"GraphDriver": {
"Data": {
"LowerDir": "/var/lib/docker/overlay2/cd7f23356b1554050c67b6824026d3f1bf16c37d41b2bae59cceff3f306ba402-init/diff:/var/lib/docker/overlay2/n21ptadgok7s9iqugjxphhbbf/diff:/var/lib/docker/overlay2/sdnptey7m1lzixt406p5gr8z3/diff:/var/lib/docker/overlay2/58f1nnq3t75qs19khgfkveb8t/diff:/var/lib/docker/overlay2/nkhdxlvbfk3kljoaw93jutdrw/diff:/var/lib/docker/overlay2/sejrlmigshuze3kg9es1odo48/diff:/var/lib/docker/overlay2/inb827fqi12cg47j7ivlda4pb/diff:/var/lib/docker/overlay2/rczyp21j590wkk53lvqbt9vbq/diff:/var/lib/docker/overlay2/k7a4rwkutdkzxloj43k0hn549/diff:/var/lib/docker/overlay2/svmvjxv49pnf0v367s1b4h41g/diff:/var/lib/docker/overlay2/a47a02bc96139d6e7300e03aed5e2501fdb7abdba258e008687825e4d8d9b466/diff:/var/lib/docker/overlay2/8856020e8fcabfcc5e15e3f074259e190d15ec67c28e8bbc6ab557822c518657/diff:/var/lib/docker/overlay2/e1c1be87b7b249b739be9c56e8f26ed7a94db966596cd663423a4af238dac30b/diff:/var/lib/docker/overlay2/72a869f3bfc1d615d412d6ba4728218cb9716545ef2385fae921041d7400b356/diff",
"MergedDir": "/var/lib/docker/overlay2/cd7f23356b1554050c67b6824026d3f1bf16c37d41b2bae59cceff3f306ba402/merged",


https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/frontend-vm?authuser=0&hl=en_US&projectNumber=44...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
"UpperDir": "/var/lib/docker/overlay2/cd7f23356b1554050c67b6824026d3f1bf16c37d41b2bae59cceff3f306ba402/diff",
"WorkDir": "/var/lib/docker/overlay2/cd7f23356b1554050c67b6824026d3f1bf16c37d41b2bae59cceff3f306ba402/work"
},
"Name": "overlay2"
},
"Mounts": [],
"Config": {
"Hostname": "3d67858f31d7",
"Domainname": "",
"User": "appuser",
"AttachStdin": false,
"AttachStdout": true,
"AttachStderr": true,
"ExposedPorts": {
"9567/tcp": {}
},
"Tty": false,
"OpenStdin": false,
"StdinOnce": false,
"Env": [
"BACKEND_URL=http://10.160.0.8:9567",
"PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
"LANG=C.UTF-8",
"GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
"PYTHON_VERSION=3.9.21",
"PYTHON_SHA256=3126f59592c9b0d798584755f2bf7b081fa1ca35ce7a6fea980108d752a05bb1"
],
"Cmd": [
"python",
"main.py"
],
"Image": "fastapi:1",
"Volumes": null,
"WorkingDir": "/app",
"Entrypoint": null,
"OnBuild": null,
"Labels": {
"com.docker.compose.config-hash": "cc01e2fa156de645ecc1fefe80efcbfec1f887b7da6ec5bc02978ec74cb4bf7b",
"com.docker.compose.container-number": "1",
"com.docker.compose.depends_on": "",
"com.docker.compose.image": "sha256:f3fe5eb6be66e207bfa473973d85616a55d1ebf637a4a1644a33427a161c37bf",
"com.docker.compose.oneoff": "False",
"com.docker.compose.project": "lab08",
"com.docker.compose.project.config_files": "/home/adiborate23/lab08/docker-compose.yml",
"com.docker.compose.project.working_dir": "/home/adiborate23/lab08",
"com.docker.compose.service": "frontend",
"com.docker.compose.version": "2.20.2"
}

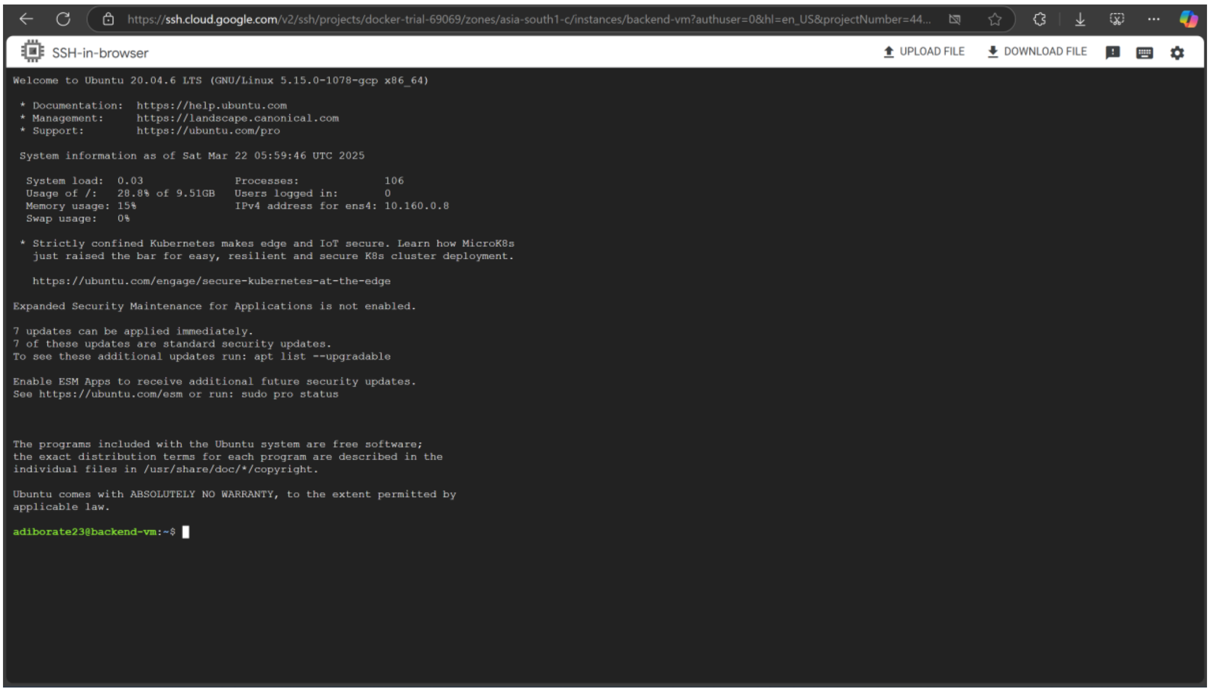


8. This is how the frontend looks like (accessed via external IP of frontend-vm)

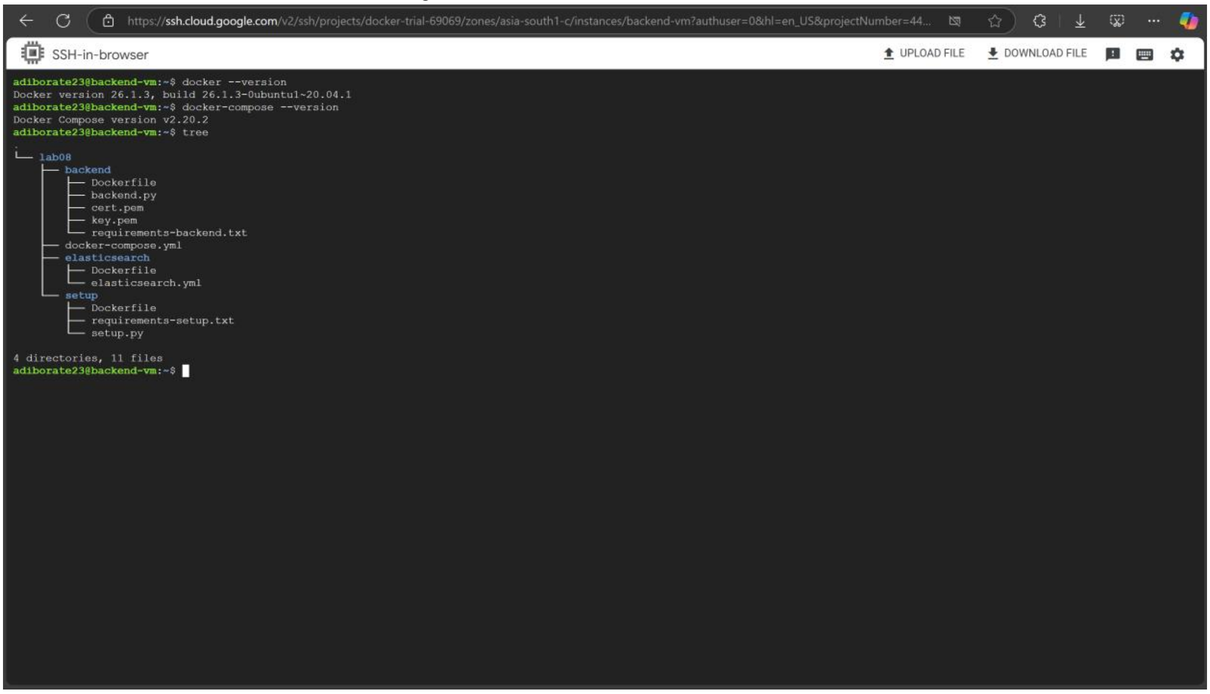


✅ Setting Up Backend

1. Running backend-vm by connecting SSH (Initial Stage):



2. After installing same required packages, and writing necessary backend files, this is how the directory structure is:



3. Dockerfile in backend directory

4. Dockerfile in elasticsearch directory

```
adiborate23@backend-vm:~/lab08/elasticsearch$ cat Dockerfile
FROM elasticsearch:7.14.0

# Add configuration for persistence
COPY elasticsearch.yml /usr/share/elasticsearch/config/elasticsearch.yml

# Set permissions
USER root
RUN chown elasticsearch:elasticsearch /usr/share/elasticsearch/config/elasticsearch.yml
USER elasticsearch

# Expose the required port used by elasticsearch
EXPOSE 9200 9300
```

5. Dockerfile in setup directory

```
adiborate23@backend-vm:~/lab08/setup$ cat Dockerfile
FROM python:3.9-alpine

WORKDIR /app

COPY requirements-setup.txt ./requirements.txt
RUN pip install --no-cache-dir -r requirements.txt

COPY setup.py ./setup.py

CMD ["python", "setup.py"]
```

✅ Building and Running the containers

1. Building and running the containers

https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...

SSH-in-browser UPLOAD FILE DOWNLOAD FILE

```
adiborate23@backend-vm:~/lab08$ docker-compose up --build -d
[+] Building 3.8s (29/29) FINISHED
 => [elasticsearch internal] load build definition from Dockerfile                0.0s
 => => transferring dockerfile: 378B                                              0.0s
 => [elasticsearch internal] load metadata for docker.io/library/elasticsearch:7.14.0   2.0s
 => [elasticsearch internal] load .dockerignore                                   0.0s
 => => transferring context: 2B                                                   0.0s
 => [elasticsearch internal] load build context                                   0.0s
 => => transferring context: 39B                                                  0.0s
 => [elasticsearch 1/3] FROM docker.io/library/elasticsearch:7.14.0@sha256:81c126e4eddbc5576285670cb3e23d7ef7892ee5e757d6d9ba870b6fe99f1219   0.0s
 => CACHED [elasticsearch 2/3] COPY elasticsearch.yml /usr/share/elasticsearch/config/elasticsearch.yml   0.0s
 => CACHED [elasticsearch 3/3] RUN chown elasticsearch:elasticsearch /usr/share/elasticsearch/config/elasticsearch.yml   0.0s
 => [elasticsearch] exporting to image                                            0.0s
 => => exporting layers                                                           0.0s
 => => writing image sha256:c950f82fcba3ed965a8b0dff7796fdebc72b956142f4fe470c81974124f0a86f   0.0s
 => => naming to docker.io/library/elasticsearch                                  0.0s
 => [setup internal] load build definition from Dockerfile                        0.0s
 => => transferring dockerfile: 229B                                              0.0s
 => [backend internal] load metadata for docker.io/library/python:3.9-alpine      1.4s
 => [backend internal] load build definition from Dockerfile                      0.0s
 => => transferring dockerfile: 453B                                              0.0s
 => [backend internal] load .dockerignore                                         0.0s
 => => transferring context: 2B                                                   0.0s
 => [setup internal] load .dockerignore                                           0.0s
 => => transferring context: 2B                                                   0.0s
 => [setup 1/5] FROM docker.io/library/python:3.9-alpine@sha256:e345f1410de8c8c40a0afac784deabce796a52f26965c41290a710d4fb47fabe   0.0s
 => [backend internal] load build context                                         0.0s
 => => transferring context: 130B                                                 0.0s
 => [setup internal] load build context                                           0.0s
 => => transferring context: 71B                                                  0.0s
 => CACHED [backend 2/5] WORKDIR /app                                             0.0s
 => CACHED [setup 3/5] COPY requirements-setup.txt ./requirements.txt             0.0s
 => CACHED [setup 4/5] RUN pip install --no-cache-dir -r requirements.txt         0.0s
 => CACHED [setup 5/5] COPY setup.py ./setup.py                                   0.0s
 => [setup] exporting to image                                                    0.0s
 => => exporting layers                                                           0.0s
 => => writing image sha256:2c14e931ae3ce747381d4677f2f7a254ec5da09bde131c0a280c8055d32b0656   0.0s
 => => naming to docker.io/library/setup                                          0.0s
 => CACHED [backend 3/9] COPY requirements-backend.txt ./requirements.txt         0.0s
 => CACHED [backend 4/9] RUN pip install --no-cache-dir -r requirements.txt       0.0s
 => CACHED [backend 5/9] COPY backend.py ./main.py                                0.0s
 => CACHED [backend 6/9] COPY key.pem .                                           0.0s
 => CACHED [backend 7/9] COPY cert.pem .                                          0.0s
 => CACHED [backend 8/9] RUN chmod 644 key.pem cert.pem                           0.0s
 => CACHED [backend 9/9] RUN adduser -D appuser                                   0.0s
 => [backend] exporting to image                                                  0.0s
 => => exporting layers                                                           0.0s
 => => writing image sha256:f5b08990990766f3179305ba2cf3dae2dfcbf4fd7f4cb8b27b0c7d275ee2ae3d   0.0s
```

```
 => => naming to docker.io/library/fastapi:2                                      0.0s
[+] Running 4/4
 ✓ Network lab08_app-network          Created                                     0.1s
 ✓ Container lab08-elasticsearch-1    Healthy                                    32.8s
 ✓ Container lab08-setup-1            Started                                    34.3s
 ✓ Container lab08-backend-1          Started                                    34.2s
adiborate23@backend-vm:~/lab08$
```

2. `docker ps` : shows all running containers

```
adiborate23@backend-vm:~/lab08$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED              STATUS                        PORTS
               NAMES
57f6337e9b87   fastapi:2         "python main.py"         About a minute ago   Up About a minute             0.0.0.0:9567->9567/tcp, :::9567->9567/tcp
               lab08-backend-1
9043d6c0c80e   elasticsearch     "/bin/tini -- /usr/l…"   About a minute ago   Up About a minute (healthy)   0.0.0.0:9200->9200/tcp, :::9200->9200/tcp, 0.0.0.0:9300->9300/tcp, :::930
0->9300/tcp    lab08-elasticsearch-1
adiborate23@backend-vm:~/lab08$
```

3. Screenshot of `docker images`

```
adiborate23@backend-vm:~/lab08$ docker images
REPOSITORY      TAG       IMAGE ID       CREATED       SIZE
fastapi         2         29ef7f2f233f   8 hours ago   73.3MB
setup           latest    2c14e931ae3c   8 hours ago   61.1MB
elasticsearch   latest    c950f82fcba3   8 hours ago   1.04GB
adiborate23@backend-vm:~/lab08$
```

At this stage, two containers are running in this backend-vm instance—

- one for the backend (`lab08-backend-1`) and

- one for Elasticsearch (`lab08-elasticsearch-1`)

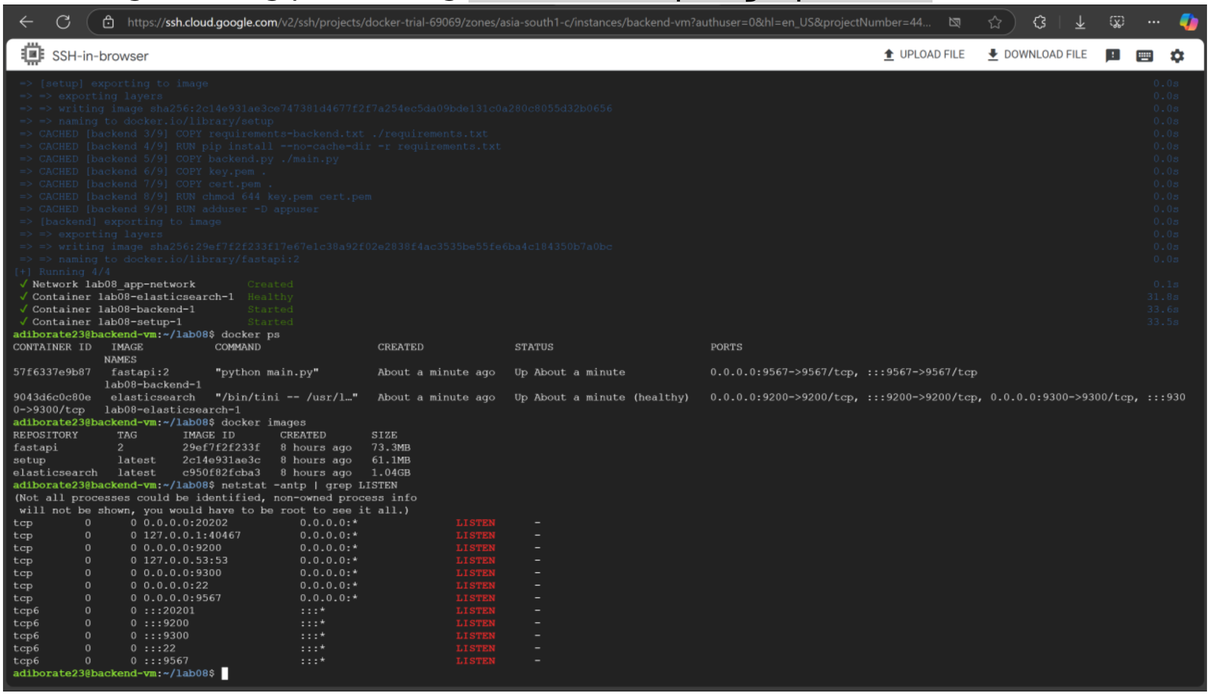
—while three images exist: `fastapi:2`, `elasticsearch`, and `setup`.

The `setup` image in `lab08-setup-1` container was designed to run only once during initialization

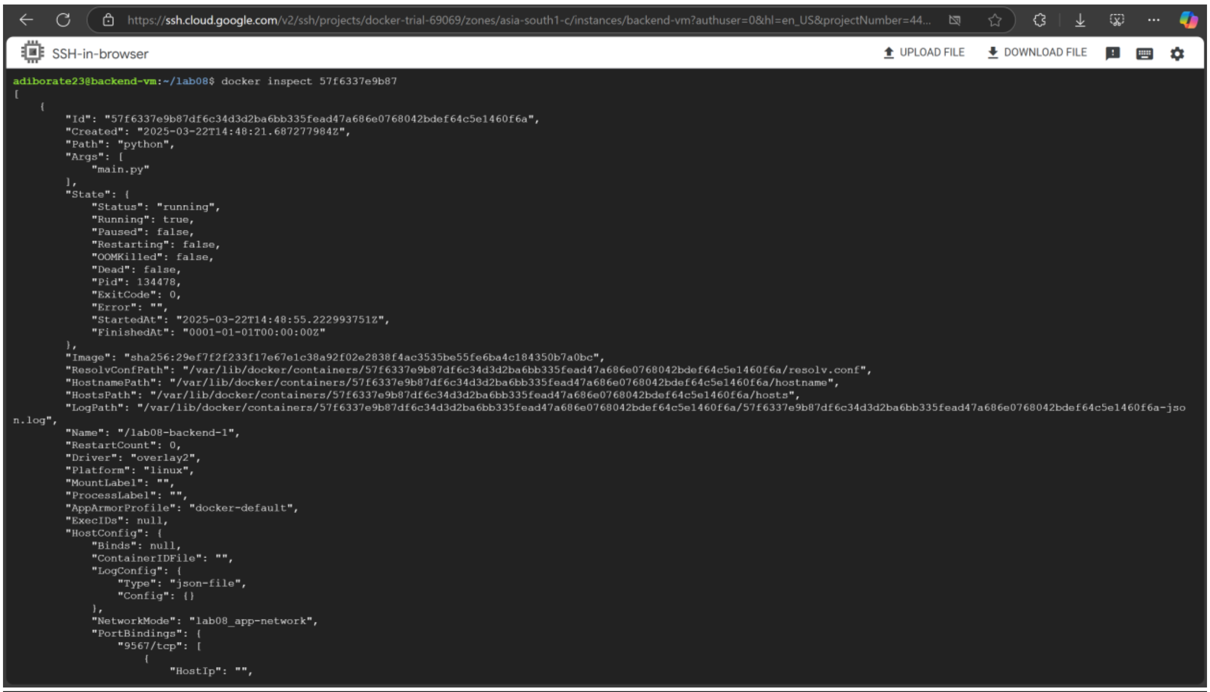
- to wait for `elasticsearch` to be healthy and up
- set up an index ("documents")
- add field mappings (id, text)
- inserts some default documents (first 4 paragraphs from https://en.wikipedia.org/wiki/India)
- and then exit.

While its container may no longer be running, the image still remains on the system for reference or potential reuse. So, the container count is less than the image count because not every image must have a continuously running container. The running containers represent active services (backend and Elasticsearch), whereas the images include everything that has been built, including one-time use components like `setup`.

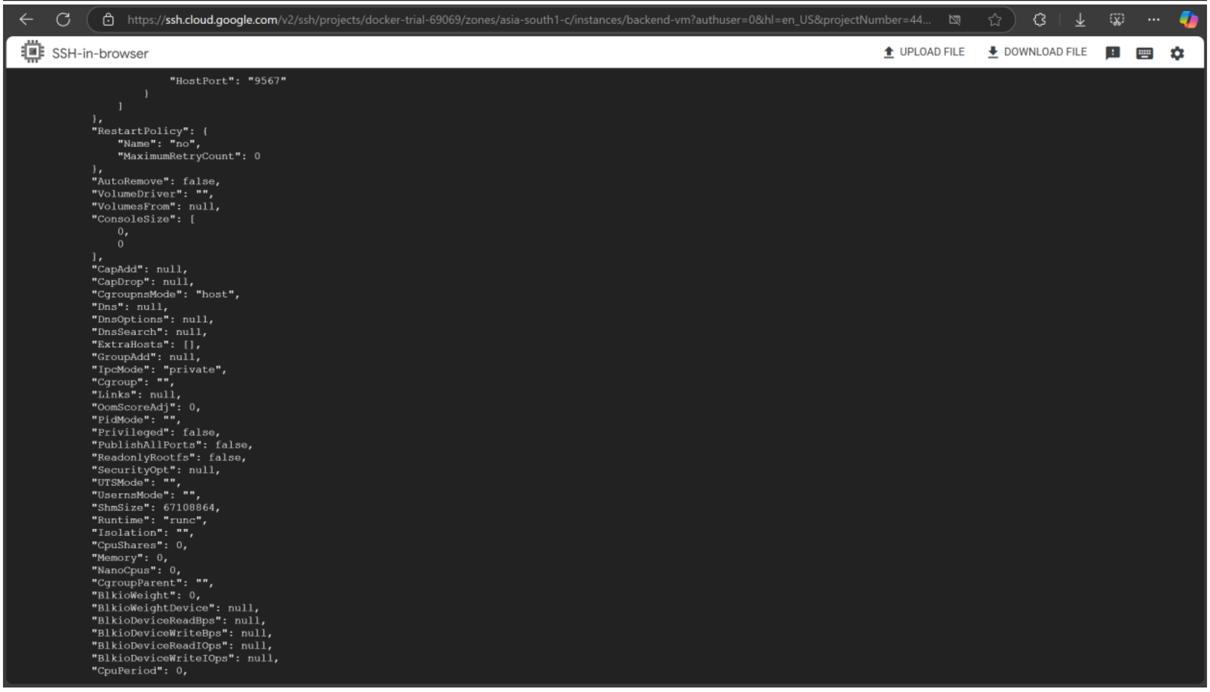
4. Checking listening ports using `netstat -antp | grep LISTEN`



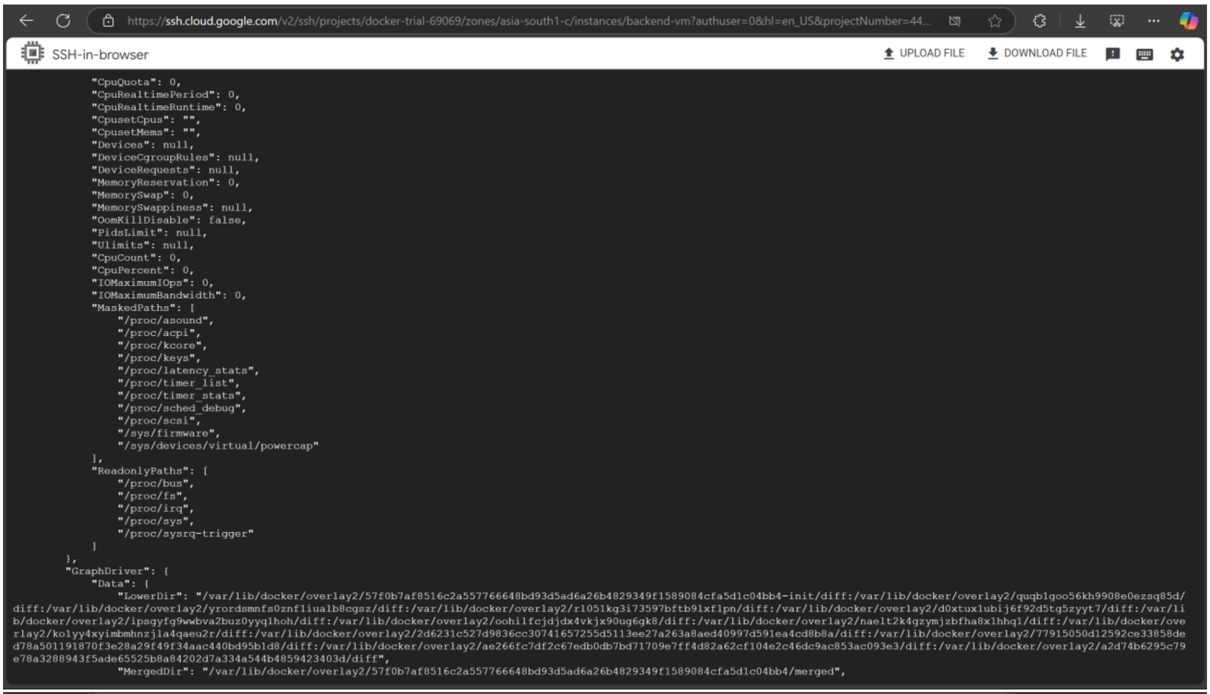
5. `docker inspect` on backend container

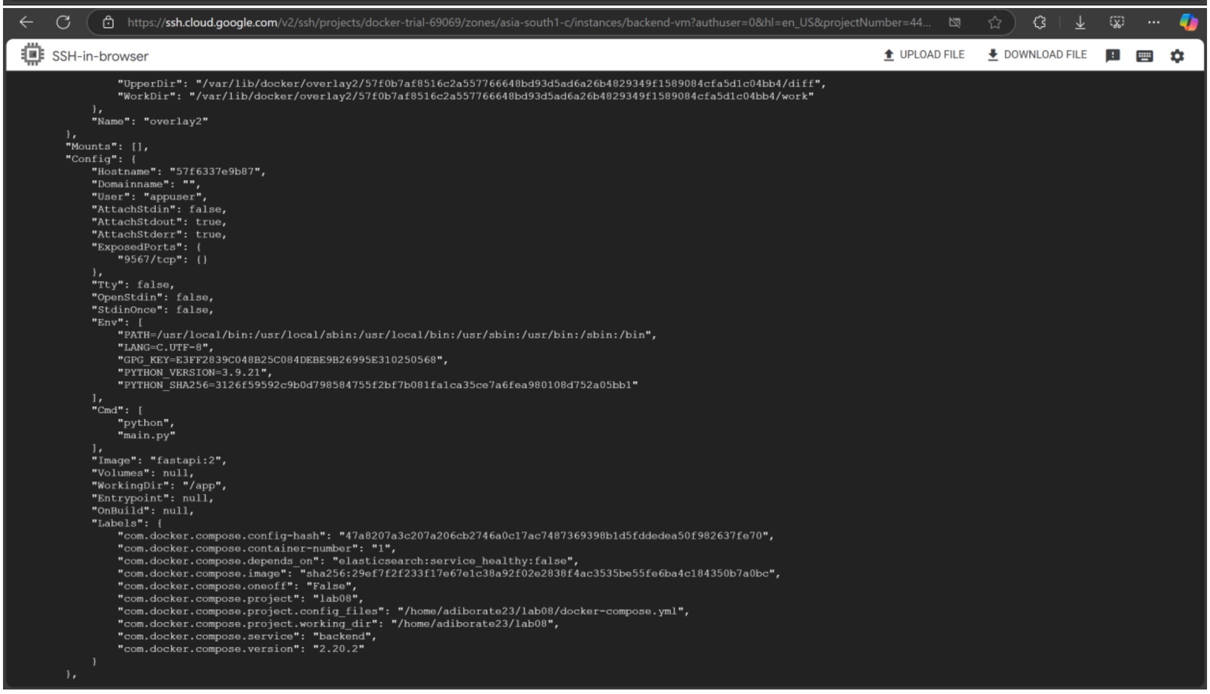


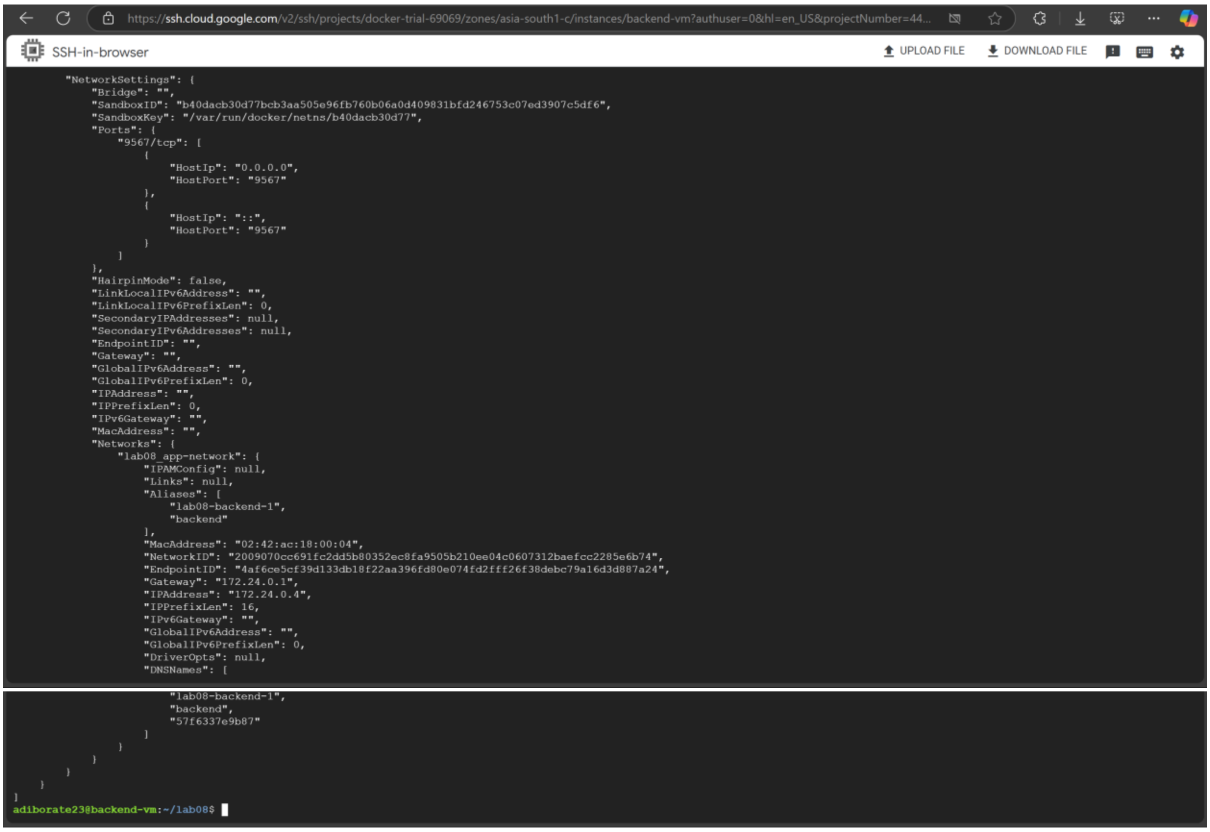

```
adiborate23@backend-vm:~/lab08$ docker inspect 57f6337e9b87
[
    {
        "Id": "57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a",
        "Created": "2025-03-22T14:48:21.687277984Z",
        "Path": "python",
        "Args": [
            "main.py"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 134478,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2025-03-22T14:48:55.222993751Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:29ef7f2f233f17e67e1c38a92f02e2838f4ac3535be55fe6ba4c184350b7a0bc",
        "ResolvConfPath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/hostname",
        "HostsPath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/hosts",
        "LogPath": "/var/lib/docker/containers/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a/57f6337e9b87df6c34d3d2ba6bb335fead47a686e0768042bdef64c5e1460f6a-jso
n.log",
        "Name": "/lab08-backend-1",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "docker-default",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "lab08_app-network",
            "PortBindings": {
                "9567/tcp": [
                    {
                        "HostIp": "",
```




```
                        "HostPort": "9567"
                    }
                ]
            },
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
            "ConsoleSize": [
                0,
                0
            ],
            "CapAdd": null,
            "CapDrop": null,
            "CgroupnsMode": "host",
            "Dns": null,
            "DnsOptions": null,
            "DnsSearch": null,
            "ExtraHosts": [],
            "GroupAdd": null,
            "IpcMode": "private",
            "Cgroup": "",
            "Links": null,
            "OomScoreAdj": 0,
            "PidMode": "",
            "Privileged": false,
            "PublishAllPorts": false,
            "ReadonlyRootfs": false,
            "SecurityOpt": null,
            "UTSMode": "",
            "UsernsMode": "",
            "ShmSize": 67108864,
            "Runtime": "runc",
            "Isolation": "",
            "CpuShares": 0,
            "Memory": 0,
            "NanoCpus": 0,
            "CgroupParent": "",
            "BlkioWeight": 0,
            "BlkioWeightDevice": null,
            "BlkioDeviceReadBps": null,
            "BlkioDeviceWriteBps": null,
            "BlkioDeviceReadIOps": null,
            "BlkioDeviceWriteIOps": null,
            "CpuPeriod": 0,
```

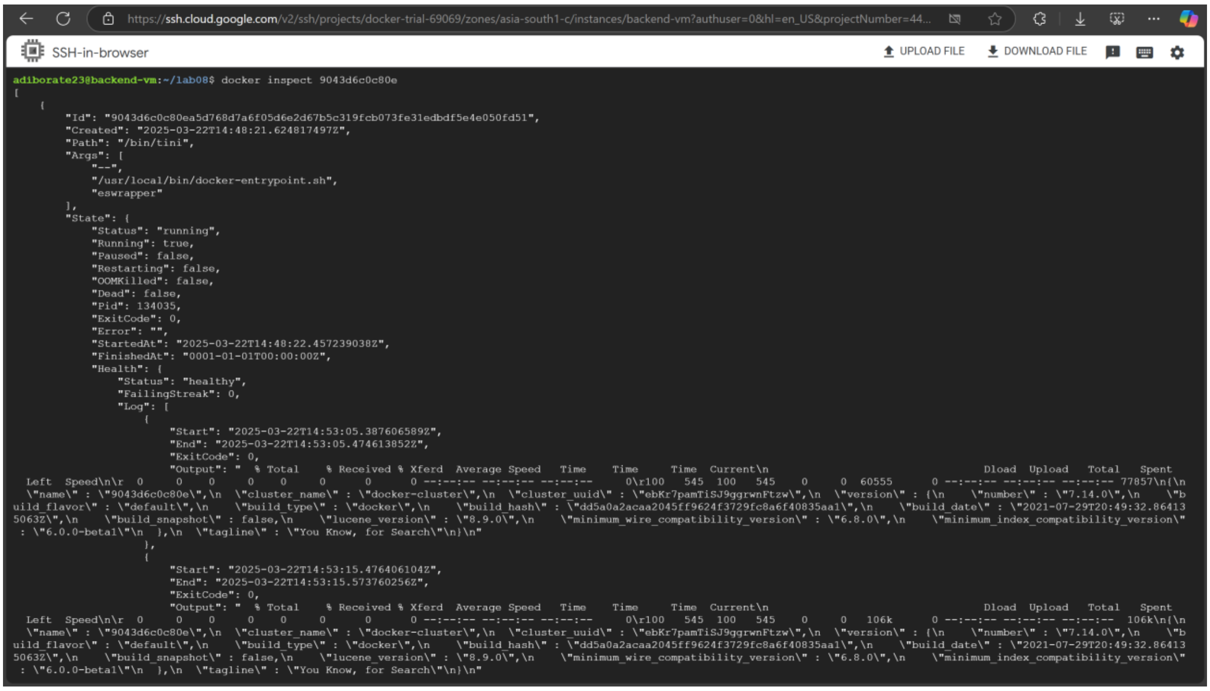

https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
"CpuQuota": 0,
"CpuRealtimePeriod": 0,
"CpuRealtimeRuntime": 0,
"CpusetCpus": "",
"CpusetMems": "",
"Devices": null,
"DeviceCgroupRules": null,
"DeviceRequests": null,
"MemoryReservation": 0,
"MemorySwap": 0,
"MemorySwappiness": null,
"OomKillDisable": false,
"PidsLimit": null,
"Ulimits": null,
"CpuCount": 0,
"CpuPercent": 0,
"IOMaximumIOps": 0,
"IOMaximumBandwidth": 0,
"MaskedPaths": [
"/proc/asound",
"/proc/acpi",
"/proc/kcore",
"/proc/keys",
"/proc/latency_stats",
"/proc/timer_list",
"/proc/timer_stats",
"/proc/sched_debug",
"/proc/scsi",
"/sys/firmware",
"/sys/devices/virtual/powercap"
],
"ReadonlyPaths": [
"/proc/bus",
"/proc/fs",
"/proc/irq",
"/proc/sys",
"/proc/sysrq-trigger"
]
},
"GraphDriver": {
"Data": {
"LowerDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4-init/diff:/var/lib/docker/overlay2/quqb1goo56kh9908e0ezsq85d/diff:/var/lib/docker/overlay2/yrordsmnfs0znf1iualb8cgsz/diff:/var/lib/docker/overlay2/r1051kg3i73597bftb91xflpn/diff:/var/lib/docker/overlay2/d0xtux1ubij6f92d5tg5zyyt7/diff:/var/lib/docker/overlay2/ipsgyfg9wwbva2buz0yyq1hoh/diff:/var/lib/docker/overlay2/oohilfcjdjdx4vkjx90ug6gk8/diff:/var/lib/docker/overlay2/naelt2k4gzymjzbfha8x1hhq1/diff:/var/lib/docker/overlay2/ko1yy4xyimbmhnzj1a4qaeu2r/diff:/var/lib/docker/overlay2/2d6231c527d9836cc30741657255d5113ee27a263a8aed40997d591ea4cd8b8a/diff:/var/lib/docker/overlay2/77915050d12592ce33858ded78a501191870f3e28a29f49f34aac440bd95b1d8/diff:/var/lib/docker/overlay2/ae266fc7df2c67edb0db7bd71709e7ff4d82a62cf104e2c46dc9ac853ac093e3/diff:/var/lib/docker/overlay2/a2d74b6295c79e78a3288943f5ade65525b8a84202d7a334a544b4859423403d/diff",
"MergedDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4/merged",


https://ssh.cloud.google.com/v2/ssh/projects/docker-trial-69069/zones/asia-south1-c/instances/backend-vm?authuser=0&hl=en_US&projectNumber=44...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
"UpperDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4/diff",
"WorkDir": "/var/lib/docker/overlay2/57f0b7af8516c2a557766648bd93d5ad6a26b4829349f1589084cfa5d1c04bb4/work"
},
"Name": "overlay2"
},
"Mounts": [],
"Config": {
"Hostname": "57f6337e9b87",
"Domainname": "",
"User": "appuser",
"AttachStdin": false,
"AttachStdout": true,
"AttachStderr": true,
"ExposedPorts": {
"9567/tcp": {}
},
"Tty": false,
"OpenStdin": false,
"StdinOnce": false,
"Env": [
"PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
"LANG=C.UTF-8",
"GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
"PYTHON_VERSION=3.9.21",
"PYTHON_SHA256=3126f59592c9b0d798584755f2bf7b081fa1ca35ce7a6fea980108d752a05bb1"
],
"Cmd": [
"python",
"main.py"
],
"Image": "fastapi:2",
"Volumes": null,
"WorkingDir": "/app",
"Entrypoint": null,
"OnBuild": null,
"Labels": {
"com.docker.compose.config-hash": "47a8207a3c207a206cb2746a0c17ac7487369398b1d5fddedea50f982637fe70",
"com.docker.compose.container-number": "1",
"com.docker.compose.depends_on": "elasticsearch:service_healthy:false",
"com.docker.compose.image": "sha256:29ef7f2f233f17e67e1c38a92f02e2838f4ac3535be55fe6ba4c184350b7a0bc",
"com.docker.compose.oneoff": "False",
"com.docker.compose.project": "lab08",
"com.docker.compose.project.config_files": "/home/adiborate23/lab08/docker-compose.yml",
"com.docker.compose.project.working_dir": "/home/adiborate23/lab08",
"com.docker.compose.service": "backend",
"com.docker.compose.version": "2.20.2"
}
},

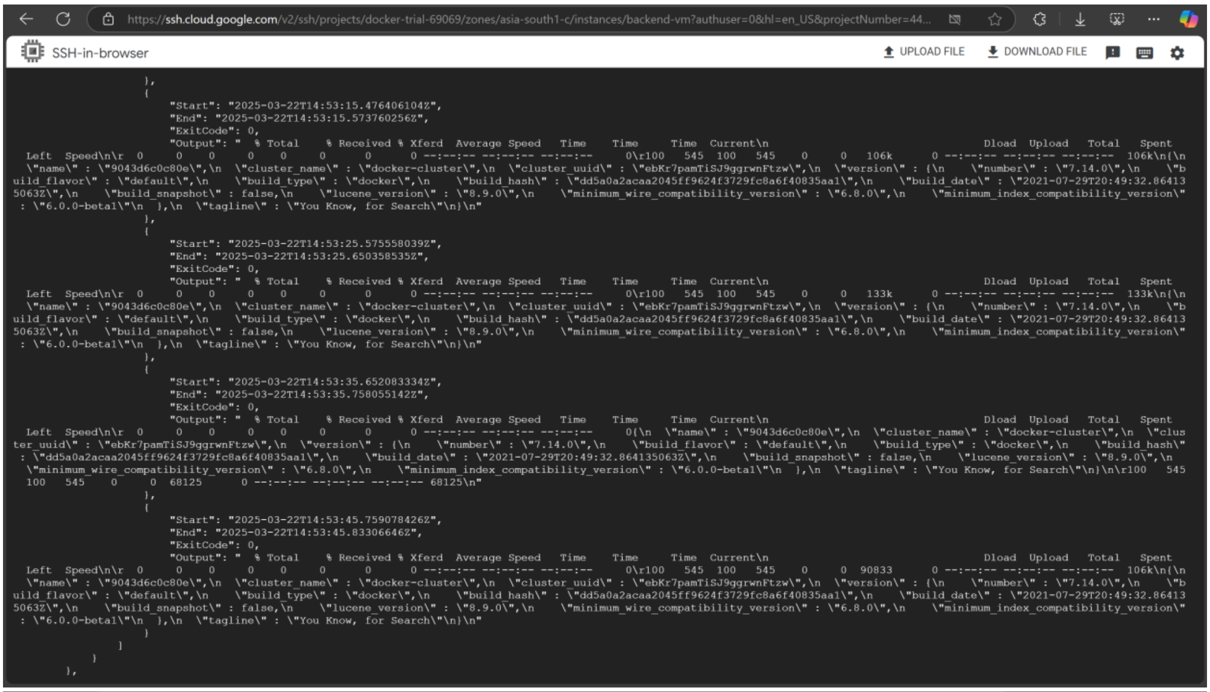



```
        "NetworkSettings": {
            "Bridge": "",
            "SandboxID": "b40dacb30d77bcb3aa505e96fb760b06a0d409831bfd246753c07ed3907c5df6",
            "SandboxKey": "/var/run/docker/netns/b40dacb30d77",
            "Ports": {
                "9567/tcp": [
                    {
                        "HostIp": "0.0.0.0",
                        "HostPort": "9567"
                    },
                    {
                        "HostIp": "::",
                        "HostPort": "9567"
                    }
                ]
            },
            "HairpinMode": false,
            "LinkLocalIPv6Address": "",
            "LinkLocalIPv6PrefixLen": 0,
            "SecondaryIPAddresses": null,
            "SecondaryIPv6Addresses": null,
            "EndpointID": "",
            "Gateway": "",
            "GlobalIPv6Address": "",
            "GlobalIPv6PrefixLen": 0,
            "IPAddress": "",
            "IPPrefixLen": 0,
            "IPv6Gateway": "",
            "MacAddress": "",
            "Networks": {
                "lab08_app-network": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": [
                        "lab08-backend-1",
                        "backend"
                    ],
                    "MacAddress": "02:42:ac:18:00:04",
                    "NetworkID": "2009070cc691fc2dd5b80352ec8fa9505b210ee04c0607312baefcc2285e6b74",
                    "EndpointID": "4af6ce5cf39d133db18f22aa396fd80e074fd2fff26f38debc79a16d3d887a24",
                    "Gateway": "172.24.0.1",
                    "IPAddress": "172.24.0.4",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DriverOpts": null,
                    "DNSNames": [
                        "lab08-backend-1",
                        "backend",
                        "57f6337e9b87"
                    ]
                }
            }
        }
    }
]
adiborate23@backend-vm:~/lab08$
```

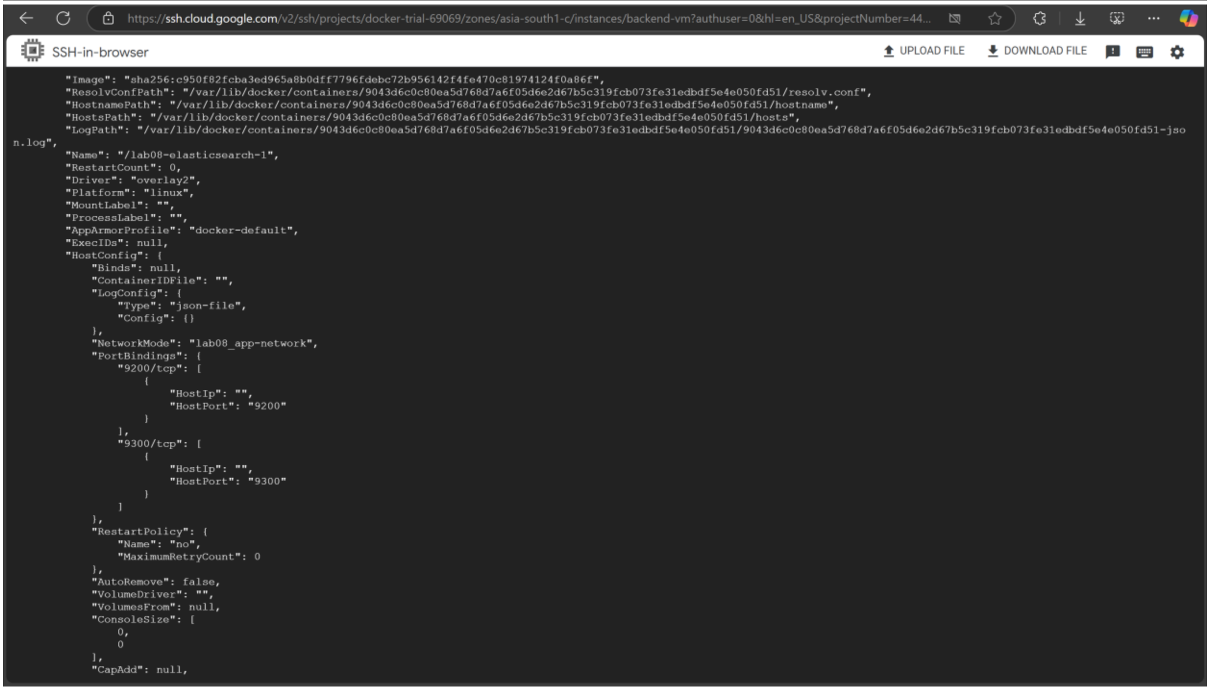
6. `docker inspect` on elasticsearch container



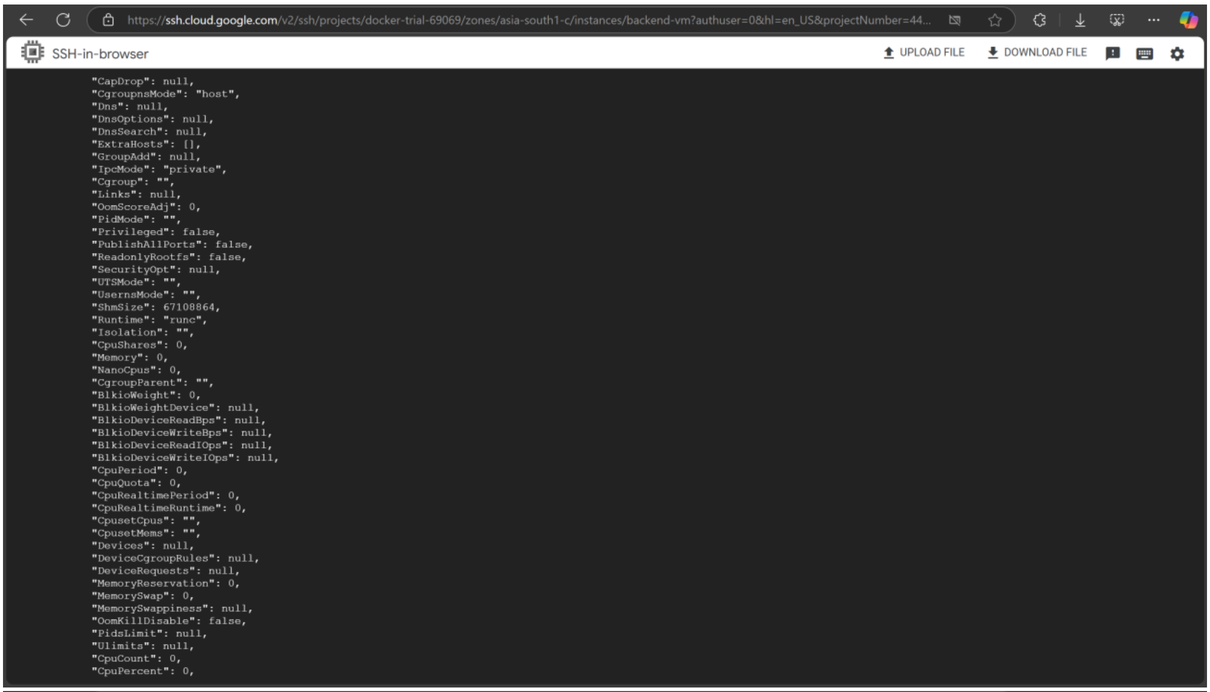

```
adiborate23@backend-vm:~/lab08$ docker inspect 9043d6c0c80e
[
    {
        "Id": "9043d6c0c80ea5d768d7a6f05d6e2d67b5c319fcb073fe31edbdf5e4e050fd51",
        "Created": "2025-03-22T14:48:21.624817497Z",
        "Path": "/bin/tini",
        "Args": [
            "--",
            "/usr/local/bin/docker-entrypoint.sh",
            "eswrapper"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 134035,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2025-03-22T14:48:22.457239038Z",
            "FinishedAt": "0001-01-01T00:00:00Z",
            "Health": {
                "Status": "healthy",
                "FailingStreak": 0,
                "Log": [
                    {
                        "Start": "2025-03-22T14:53:05.387606589Z",
                        "End": "2025-03-22T14:53:05.474613852Z",
                        "ExitCode": 0,
                        "Output": "  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current\n                                 Dload  Upload   Total   Spent
  Left  Speed\n\r  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0\r100   545  100   545    0     0  60555      0 --:--:-- --:--:-- --:--:-- 77857\n{\n
  \"name\" : \"9043d6c0c80e\",\n  \"cluster_name\" : \"docker-cluster\",\n  \"cluster_uuid\" : \"ebKr7pamTiSJ9ggrwnFtzw\",\n  \"version\" : {\n    \"number\" : \"7.14.0\",\n    \"b
uild_flavor\" : \"default\",\n    \"build_type\" : \"docker\",\n    \"build_hash\" : \"dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1\",\n    \"build_date\" : \"2021-07-29T20:49:32.86413
5063Z\",\n    \"build_snapshot\" : false,\n    \"lucene_version\" : \"8.9.0\",\n    \"minimum_wire_compatibility_version\" : \"6.8.0\",\n    \"minimum_index_compatibility_version\"
 : \"6.0.0-beta1\"\n  },\n  \"tagline\" : \"You Know, for Search\"\n}\n"
                    },
                    {
                        "Start": "2025-03-22T14:53:15.476406104Z",
                        "End": "2025-03-22T14:53:15.573760256Z",
                        "ExitCode": 0,
                        "Output": "  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current\n                                 Dload  Upload   Total   Spent
  Left  Speed\n\r  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0\r100   545  100   545    0     0   106k      0 --:--:-- --:--:-- --:--:--  106k\n{\n
  \"name\" : \"9043d6c0c80e\",\n  \"cluster_name\" : \"docker-cluster\",\n  \"cluster_uuid\" : \"ebKr7pamTiSJ9ggrwnFtzw\",\n  \"version\" : {\n    \"number\" : \"7.14.0\",\n    \"b
uild_flavor\" : \"default\",\n    \"build_type\" : \"docker\",\n    \"build_hash\" : \"dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1\",\n    \"build_date\" : \"2021-07-29T20:49:32.86413
5063Z\",\n    \"build_snapshot\" : false,\n    \"lucene_version\" : \"8.9.0\",\n    \"minimum_wire_compatibility_version\" : \"6.8.0\",\n    \"minimum_index_compatibility_version\"
 : \"6.0.0-beta1\"\n  },\n  \"tagline\" : \"You Know, for Search\"\n}\n"
```



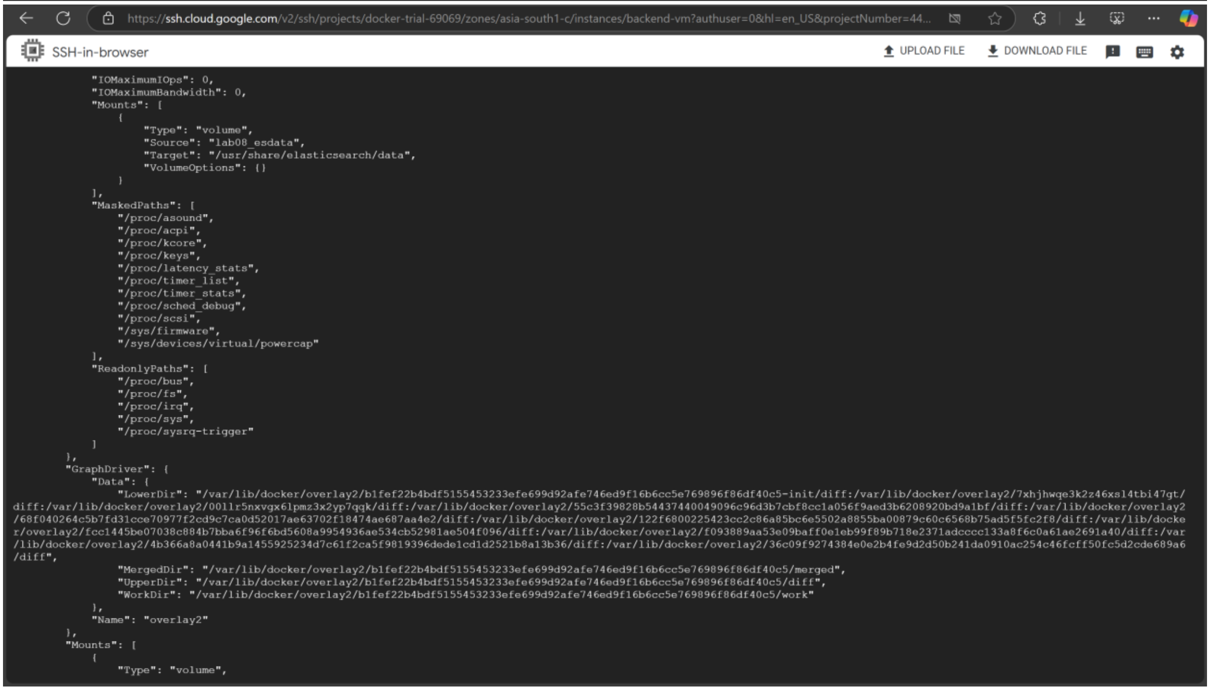

```
                },
                {
                    "Start": "2025-03-22T14:53:15.476406104Z",
                    "End": "2025-03-22T14:53:15.573760256Z",
                    "ExitCode": 0,
                    "Output": "  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current\n                                 Dload  Upload   Total   Spent    Left  Speed\n\r  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0\r100   545  100   545    0     0   106k      0 --:--:-- --:--:-- --:--:--  106k\n{\n  \"name\" : \"9043d6c0c80e\",\n  \"cluster_name\" : \"docker-cluster\",\n  \"cluster_uuid\" : \"ebKr7pamTiSJ9ggrwnFtzw\",\n  \"version\" : {\n    \"number\" : \"7.14.0\",\n    \"build_flavor\" : \"default\",\n    \"build_type\" : \"docker\",\n    \"build_hash\" : \"dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1\",\n    \"build_date\" : \"2021-07-29T20:49:32.864135063Z\",\n    \"build_snapshot\" : false,\n    \"lucene_version\" : \"8.9.0\",\n    \"minimum_wire_compatibility_version\" : \"6.8.0\",\n    \"minimum_index_compatibility_version\" : \"6.0.0-beta1\"\n  },\n  \"tagline\" : \"You Know, for Search\"\n}\n"
                },
                {
                    "Start": "2025-03-22T14:53:25.575558039Z",
                    "End": "2025-03-22T14:53:25.650358535Z",
                    "ExitCode": 0,
                    "Output": "  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current\n                                 Dload  Upload   Total   Spent    Left  Speed\n\r  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0\r100   545  100   545    0     0   133k      0 --:--:-- --:--:-- --:--:--  133k\n{\n  \"name\" : \"9043d6c0c80e\",\n  \"cluster_name\" : \"docker-cluster\",\n  \"cluster_uuid\" : \"ebKr7pamTiSJ9ggrwnFtzw\",\n  \"version\" : {\n    \"number\" : \"7.14.0\",\n    \"build_flavor\" : \"default\",\n    \"build_type\" : \"docker\",\n    \"build_hash\" : \"dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1\",\n    \"build_date\" : \"2021-07-29T20:49:32.864135063Z\",\n    \"build_snapshot\" : false,\n    \"lucene_version\" : \"8.9.0\",\n    \"minimum_wire_compatibility_version\" : \"6.8.0\",\n    \"minimum_index_compatibility_version\" : \"6.0.0-beta1\"\n  },\n  \"tagline\" : \"You Know, for Search\"\n}\n"
                },
                {
                    "Start": "2025-03-22T14:53:35.652083334Z",
                    "End": "2025-03-22T14:53:35.758055142Z",
                    "ExitCode": 0,
                    "Output": "  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current\n                                 Dload  Upload   Total   Spent    Left  Speed\n\r  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0{\n  \"name\" : \"9043d6c0c80e\",\n  \"cluster_name\" : \"docker-cluster\",\n  \"cluster_uuid\" : \"ebKr7pamTiSJ9ggrwnFtzw\",\n  \"version\" : {\n    \"number\" : \"7.14.0\",\n    \"build_flavor\" : \"default\",\n    \"build_type\" : \"docker\",\n    \"build_hash\" : \"dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1\",\n    \"build_date\" : \"2021-07-29T20:49:32.864135063Z\",\n    \"build_snapshot\" : false,\n    \"lucene_version\" : \"8.9.0\",\n    \"minimum_wire_compatibility_version\" : \"6.8.0\",\n    \"minimum_index_compatibility_version\" : \"6.0.0-beta1\"\n  },\n  \"tagline\" : \"You Know, for Search\"\n}\n\r100   545  100   545    0     0  68125      0 --:--:-- --:--:-- --:--:-- 68125\n"
                },
                {
                    "Start": "2025-03-22T14:53:45.759078426Z",
                    "End": "2025-03-22T14:53:45.833066646Z",
                    "ExitCode": 0,
                    "Output": "  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current\n                                 Dload  Upload   Total   Spent    Left  Speed\n\r  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0\r100   545  100   545    0     0  90833      0 --:--:-- --:--:-- --:--:--  106k\n{\n  \"name\" : \"9043d6c0c80e\",\n  \"cluster_name\" : \"docker-cluster\",\n  \"cluster_uuid\" : \"ebKr7pamTiSJ9ggrwnFtzw\",\n  \"version\" : {\n    \"number\" : \"7.14.0\",\n    \"build_flavor\" : \"default\",\n    \"build_type\" : \"docker\",\n    \"build_hash\" : \"dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1\",\n    \"build_date\" : \"2021-07-29T20:49:32.864135063Z\",\n    \"build_snapshot\" : false,\n    \"lucene_version\" : \"8.9.0\",\n    \"minimum_wire_compatibility_version\" : \"6.8.0\",\n    \"minimum_index_compatibility_version\" : \"6.0.0-beta1\"\n  },\n  \"tagline\" : \"You Know, for Search\"\n}\n"
                }
            ]
        }
    },
```


```
        "Image": "sha256:c950f82fcba3ed965a8b0dff7796fdebc72b956142f4fe470c81974124f0a86f",
        "ResolvConfPath": "/var/lib/docker/containers/9043d6c0c80ea5d768d7a6f05d6e2d67b5c319fcb073fe31edbdf5e4e050fd51/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/9043d6c0c80ea5d768d7a6f05d6e2d67b5c319fcb073fe31edbdf5e4e050fd51/hostname",
        "HostsPath": "/var/lib/docker/containers/9043d6c0c80ea5d768d7a6f05d6e2d67b5c319fcb073fe31edbdf5e4e050fd51/hosts",
        "LogPath": "/var/lib/docker/containers/9043d6c0c80ea5d768d7a6f05d6e2d67b5c319fcb073fe31edbdf5e4e050fd51/9043d6c0c80ea5d768d7a6f05d6e2d67b5c319fcb073fe31edbdf5e4e050fd51-json.log",
        "Name": "/lab08-elasticsearch-1",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "docker-default",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "lab08_app-network",
            "PortBindings": {
                "9200/tcp": [
                    {
                        "HostIp": "",
                        "HostPort": "9200"
                    }
                ],
                "9300/tcp": [
                    {
                        "HostIp": "",
                        "HostPort": "9300"
                    }
                ]
            },
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
            "ConsoleSize": [
                0,
                0
            ],
            "CapAdd": null,
```



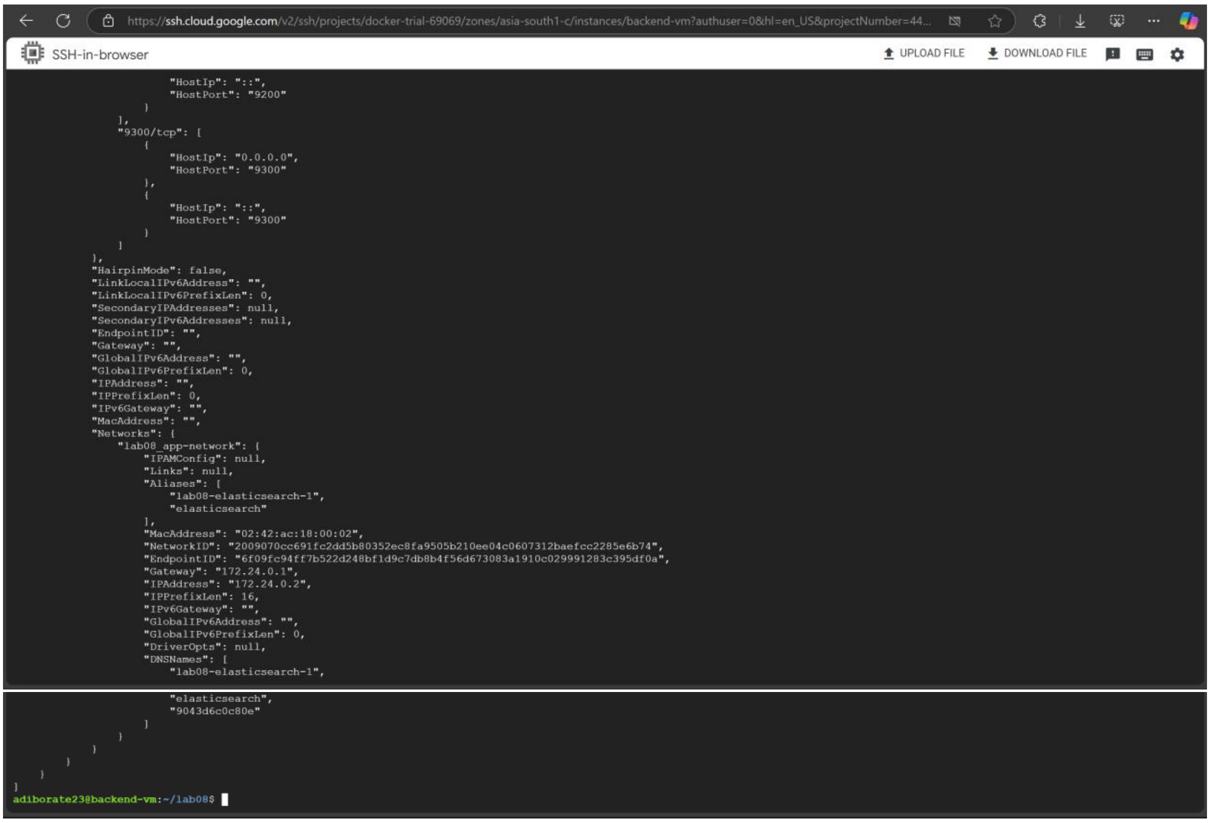

```
            "CapDrop": null,
            "CgroupnsMode": "host",
            "Dns": null,
            "DnsOptions": null,
            "DnsSearch": null,
            "ExtraHosts": [],
            "GroupAdd": null,
            "IpcMode": "private",
            "Cgroup": "",
            "Links": null,
            "OomScoreAdj": 0,
            "PidMode": "",
            "Privileged": false,
            "PublishAllPorts": false,
            "ReadonlyRootfs": false,
            "SecurityOpt": null,
            "UTSMode": "",
            "UsernsMode": "",
            "ShmSize": 67108864,
            "Runtime": "runc",
            "Isolation": "",
            "CpuShares": 0,
            "Memory": 0,
            "NanoCpus": 0,
            "CgroupParent": "",
            "BlkioWeight": 0,
            "BlkioWeightDevice": null,
            "BlkioDeviceReadBps": null,
            "BlkioDeviceWriteBps": null,
            "BlkioDeviceReadIOps": null,
            "BlkioDeviceWriteIOps": null,
            "CpuPeriod": 0,
            "CpuQuota": 0,
            "CpuRealtimePeriod": 0,
            "CpuRealtimeRuntime": 0,
            "CpusetCpus": "",
            "CpusetMems": "",
            "Devices": null,
            "DeviceCgroupRules": null,
            "DeviceRequests": null,
            "MemoryReservation": 0,
            "MemorySwap": 0,
            "MemorySwappiness": null,
            "OomKillDisable": false,
            "PidsLimit": null,
            "Ulimits": null,
            "CpuCount": 0,
            "CpuPercent": 0,
```


```
            "IOMaximumIOps": 0,
            "IOMaximumBandwidth": 0,
            "Mounts": [
                {
                    "Type": "volume",
                    "Source": "lab08_esdata",
                    "Target": "/usr/share/elasticsearch/data",
                    "VolumeOptions": {}
                }
            ],
            "MaskedPaths": [
                "/proc/asound",
                "/proc/acpi",
                "/proc/kcore",
                "/proc/keys",
                "/proc/latency_stats",
                "/proc/timer_list",
                "/proc/timer_stats",
                "/proc/sched_debug",
                "/proc/scsi",
                "/sys/firmware",
                "/sys/devices/virtual/powercap"
            ],
            "ReadonlyPaths": [
                "/proc/bus",
                "/proc/fs",
                "/proc/irq",
                "/proc/sys",
                "/proc/sysrq-trigger"
            ]
        },
        "GraphDriver": {
            "Data": {
                "LowerDir": "/var/lib/docker/overlay2/b1fef22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5-init/diff:/var/lib/docker/overlay2/7xhjhwqe3k2z46xsl4tbi47gt/
diff:/var/lib/docker/overlay2/00llr5nxvgx6lpmz3x2yp7qqk/diff:/var/lib/docker/overlay2/55c3f39828b54437440049096c96d3b7cbf8cc1a056f9aed3b6208920bd9a1bf/diff:/var/lib/docker/overlay2
/68f040264c5b7fd31cce70977f2cd9c7ca0d52017ae63702f18474ae687aa4e2/diff:/var/lib/docker/overlay2/122f6800225423cc2c86a85bc6e5502a8855ba00879c60c6568b75ad5f5fc2f8/diff:/var/lib/docke
r/overlay2/fcc1445be07038c884b7bba6f96f6bd5608a9954936ae534cb52981ae504f096/diff:/var/lib/docker/overlay2/f093889aa53e09baff0e1eb99f89b718e2371adcccc133a8f6c0a61ae2691a40/diff:/var
/lib/docker/overlay2/4b366a8a0441b9a1455925234d7c61f2ca5f9819396dede1cd1d2521b8a13b36/diff:/var/lib/docker/overlay2/36c09f9274384e0e2b4fe9d2d50b241da0910ac254c46fcff50fc5d2cde689a6
/diff",
                "MergedDir": "/var/lib/docker/overlay2/b1fef22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5/merged",
                "UpperDir": "/var/lib/docker/overlay2/b1fef22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5/diff",
                "WorkDir": "/var/lib/docker/overlay2/b1fef22b4bdf5155453233efe699d92afe746ed9f16b6cc5e769896f86df40c5/work"
            },
            "Name": "overlay2"
        },
        "Mounts": [
            {
                "Type": "volume",
```

```
            "Name": "lab08_esdata",
            "Source": "/var/lib/docker/volumes/lab08_esdata/_data",
            "Destination": "/usr/share/elasticsearch/data",
            "Driver": "local",
            "Mode": "z",
            "RW": true,
            "Propagation": ""
        }
    ],
    "Config": {
        "Hostname": "9043d6c0c80e",
        "Domainname": "",
        "User": "elasticsearch",
        "AttachStdin": false,
        "AttachStdout": true,
        "AttachStderr": true,
        "ExposedPorts": {
            "9200/tcp": {},
            "9300/tcp": {}
        },
        "Tty": false,
        "OpenStdin": false,
        "StdinOnce": false,
        "Env": [
            "discovery.type=single-node",
            "ES_JAVA_OPTS=-Xms512m -Xmx512m",
            "PATH=/usr/share/elasticsearch/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
            "ELASTIC_CONTAINER=true"
        ],
        "Cmd": [
            "eswrapper"
        ],
        "Healthcheck": {
            "Test": [
                "CMD",
                "curl",
                "-f",
                "http://localhost:9200"
            ],
            "Interval": 10000000000,
            "Timeout": 5000000000,
            "StartPeriod": 20000000000,
            "Retries": 5
        },
        "Image": "elasticsearch",
        "Volumes": null,
        "WorkingDir": "/usr/share/elasticsearch",
        "Entrypoint": [
```

```
            "/bin/tini",
            "--",
            "/usr/local/bin/docker-entrypoint.sh"
        ],
        "OnBuild": null,
        "Labels": {
            "com.docker.compose.config-hash": "d37824d669286c91a4c612e2264ba054289988f663217bd17e96cc0ca3194c59",
            "com.docker.compose.container-number": "1",
            "com.docker.compose.depends_on": "",
            "com.docker.compose.image": "sha256:c950f82fcba3ed965a8b0dff7796fdebc72b956142f4fe470c81974124f0a86f",
            "com.docker.compose.oneoff": "False",
            "com.docker.compose.project": "lab08",
            "com.docker.compose.project.config_files": "/home/adiborate23/lab08/docker-compose.yml",
            "com.docker.compose.project.working_dir": "/home/adiborate23/lab08",
            "com.docker.compose.service": "elasticsearch",
            "com.docker.compose.version": "2.20.2",
            "org.label-schema.build-date": "2021-07-29T20:49:32.864135063Z",
            "org.label-schema.license": "Elastic-License-2.0",
            "org.label-schema.name": "Elasticsearch",
            "org.label-schema.schema-version": "1.0",
            "org.label-schema.url": "https://www.elastic.co/products/elasticsearch",
            "org.label-schema.usage": "https://www.elastic.co/guide/en/elasticsearch/reference/index.html",
            "org.label-schema.vcs-ref": "dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1",
            "org.label-schema.vcs-url": "https://github.com/elastic/elasticsearch",
            "org.label-schema.vendor": "Elastic",
            "org.label-schema.version": "7.14.0",
            "org.opencontainers.image.created": "2021-07-29T20:49:32.864135063Z",
            "org.opencontainers.image.documentation": "https://www.elastic.co/guide/en/elasticsearch/reference/index.html",
            "org.opencontainers.image.licenses": "Elastic-License-2.0",
            "org.opencontainers.image.revision": "dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1",
            "org.opencontainers.image.source": "https://github.com/elastic/elasticsearch",
            "org.opencontainers.image.title": "Elasticsearch",
            "org.opencontainers.image.url": "https://www.elastic.co/products/elasticsearch",
            "org.opencontainers.image.vendor": "Elastic",
            "org.opencontainers.image.version": "7.14.0"
        }
    },
    "NetworkSettings": {
        "Bridge": "",
        "SandboxID": "4f692be57d90aa1762226e08f57aab484f3ea113b9af5b4d6b2014a6ec9aeb29",
        "SandboxKey": "/var/run/docker/netns/4f692be57d90",
        "Ports": {
            "9200/tcp": [
                {
                    "HostIp": "0.0.0.0",
                    "HostPort": "9200"
                },
                {
```



✅ Running FastAPI and Elasticsearch on Specific Ports

1. FastAPI (Frontend and Backend)
   Both FastAPI services (frontend and backend) are explicitly configured to run on port `9567`
   - In `main.py` (Frontend):

```
if __name__ == "__main__":
    uvicorn.run(
        app,
        host="0.0.0.0",
        port=9567,
        ssl_keyfile="key.pem",
        ssl_certfile="cert.pem"
    )
```

   - In `backend.py` (Backend):

```
# Run the FastAPI app
if __name__ == "__main__":
    uvicorn.run(app, host="0.0.0.0", port=9567)
```

   - Dockerfile exposes the port:

```
    EXPOSE 9567
```

   - Docker Compose maps the container port to host:

```
    ports:
      - "9567:9567"
```

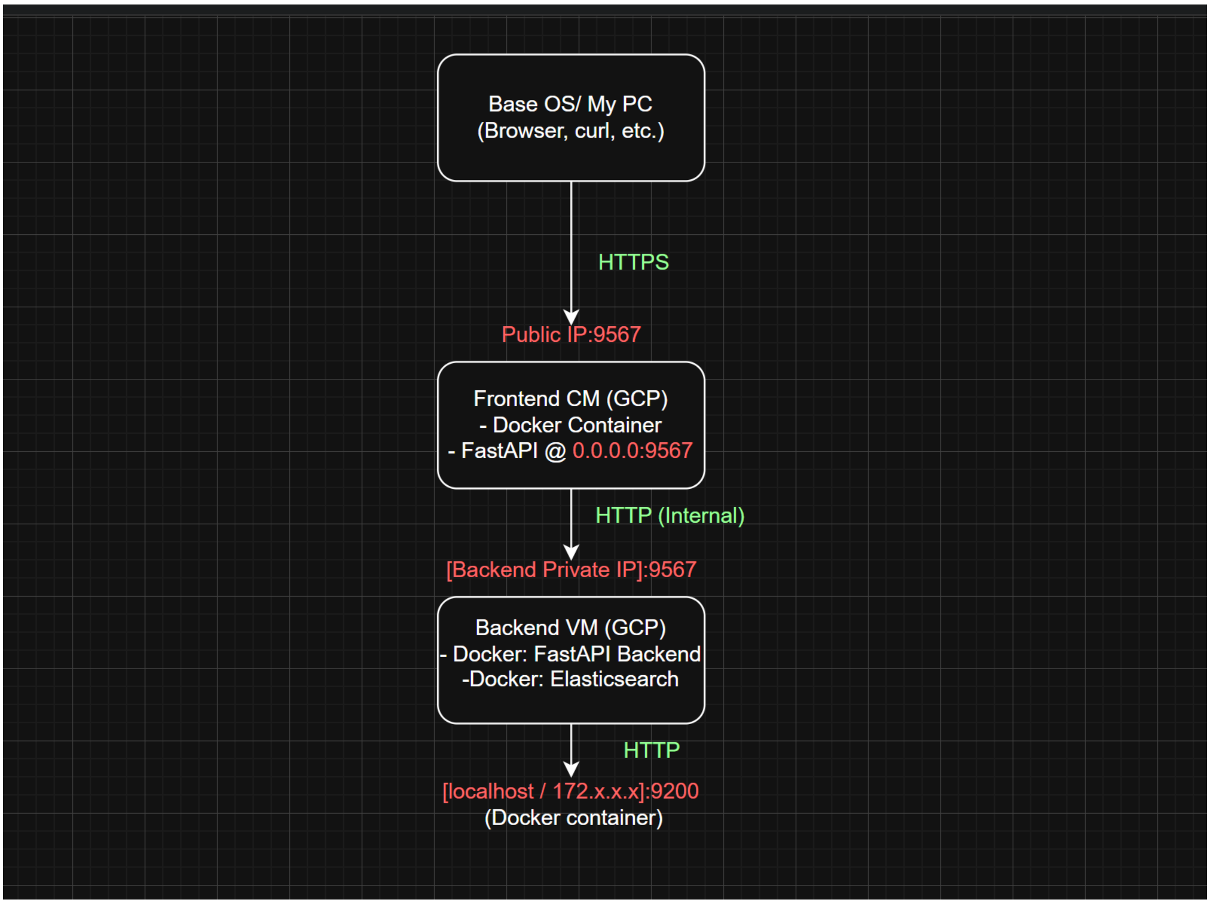
2. Elasticsearch (Private Access Only)
   Elasticsearch runs on standard ports internally: 9200 (for REST API) and 9300 (for node communication), but they are not published to the host machine. This is done to :
   - Prevent external or frontend access (even via private IP)
   - Limit Elasticsearch access to the backend container only
   - Improve security by reducing attack surface

   Docker Compose (Backend VM):

```
    # Removed public port bindings to restrict access
    # ports:
    #   - "9200:9200"
    #   - "9300:9300"
```

   Elasticsearch still works fine because the backend and Elasticsearch containers are on the same internal Docker network (`app-network`), allowing internal communication.

✅ Check the accessibility of ports



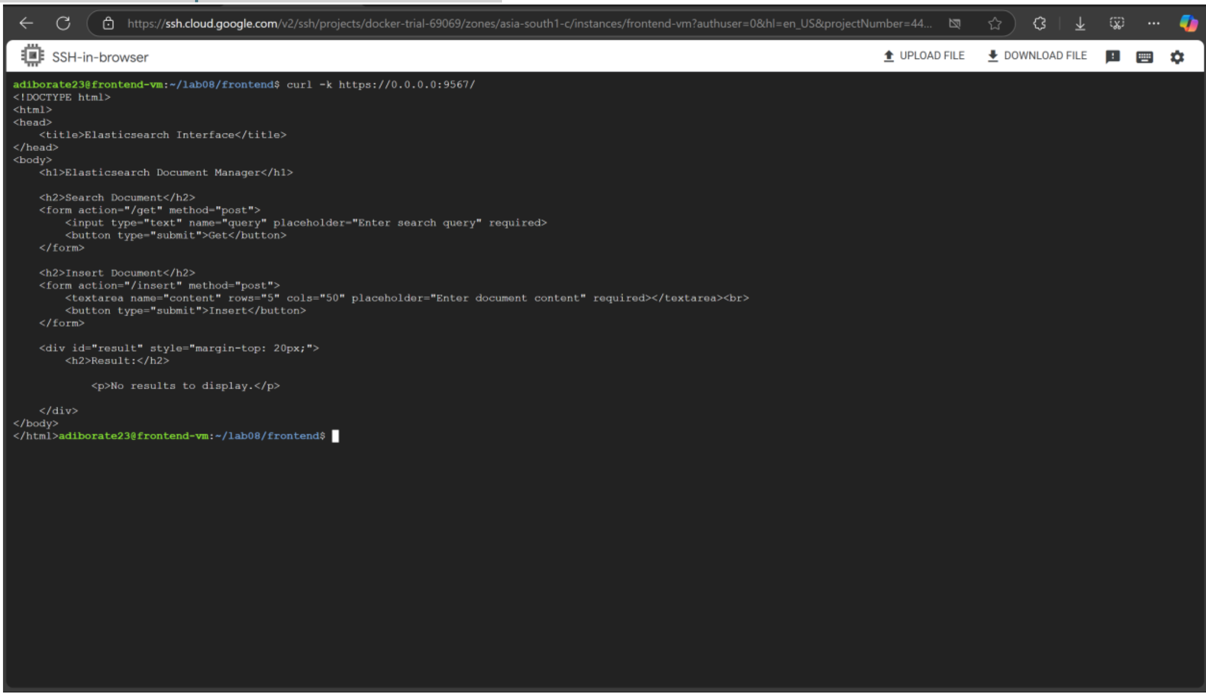
1. Inside frontend-vm:

   a. `curl http://10.160.0.8:9567/docs`

```
adiborate23@frontend-vm:~$ curl http://10.160.0.8:9567/docs

    <!DOCTYPE html>
    <html>
    <head>
    <link type="text/css" rel="stylesheet" href="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui.css">
    <link rel="shortcut icon" href="https://fastapi.tiangolo.com/img/favicon.png">
    <title>FastAPI - Swagger UI</title>
    </head>
    <body>
    <div id="swagger-ui">
    </div>
    <script src="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui-bundle.js"></script>
    <!-- `SwaggerUIBundle` is now available on the page -->
    <script>
    const ui = SwaggerUIBundle({
        url: '/openapi.json',
    "dom_id": "#swagger-ui",
"layout": "BaseLayout",
"deepLinking": true,
"showExtensions": true,
"showCommonExtensions": true,
oauth2RedirectUrl: window.location.origin + '/docs/oauth2-redirect',
    presets: [
        SwaggerUIBundle.presets.apis,
        SwaggerUIBundle.SwaggerUIStandalonePreset
        ],
    })
    </script>
    </body>
    </html>
    adiborate23@frontend-vm:~$
```

   --WORKED: Shows frontend can reach backend via internal IP. This confirms inter-VM internal networking is correctly set up.

   b. `curl -k https://0.0.0.0:9567/`



```
adiborate23@frontend-vm:~/lab08/frontend$ curl -k https://0.0.0.0:9567/
<!DOCTYPE html>
<html>
<head>
    <title>Elasticsearch Interface</title>
</head>
<body>
    <h1>Elasticsearch Document Manager</h1>

    <h2>Search Document</h2>
    <form action="/get" method="post">
        <input type="text" name="query" placeholder="Enter search query" required>
        <button type="submit">Get</button>
    </form>

    <h2>Insert Document</h2>
    <form action="/insert" method="post">
        <textarea name="content" rows="5" cols="50" placeholder="Enter document content" required></textarea><br>
        <button type="submit">Insert</button>
    </form>

    <div id="result" style="margin-top: 20px;">
        <h2>Result:</h2>

            <p>No results to display.</p>

    </div>
</body>
</html>adiborate23@frontend-vm:~/lab08/frontend$
```

   --WORKED: Confirms FastAPI in frontend is bound to all interfaces.

   c.
```
curl -k -X POST https://34.47.186.152:9567/insert \
--data-urlencode "content=test from frontend"
```

```
adiborate23@frontend-vm:~/lab08$ curl -k -X POST https://34.47.186.152:9567/insert \
> --data-urlencode "content=test from frontend"
<!DOCTYPE html>
<html>
<head>
    <title>Elasticsearch Interface</title>
</head>
<body>
    <h1>Elasticsearch Document Manager</h1>

    <h2>Search Document</h2>
    <form action="/get" method="post">
        <input type="text" name="query" placeholder="Enter search query" required>
        <button type="submit">Get</button>
    </form>

    <h2>Insert Document</h2>
    <form action="/insert" method="post">
        <textarea name="content" rows="5" cols="50" placeholder="Enter document content" required></textarea><br>
        <button type="submit">Insert</button>
    </form>

    <div id="result" style="margin-top: 20px;">
        <h2>Result:</h2>


                <p>Document inserted successfully with ID: 9tp4v5UBylE19jowNAd0</p>


    </div>
</body>
</html>adiborate23@frontend-vm:~/lab08$
```

--WORKED: Confirms public access to frontend is functional.

d. `curl -v --connect-timeout 5 http://34.47.206.180:9567`

```
adiborate23@frontend-vm:~/lab08/frontend$ curl -v --connect-timeout 5 http://34.47.206.180:9567
*   Trying 34.47.206.180:9567...
* TCP_NODELAY set
* Connection timed out after 5000 milliseconds
* Closing connection 0
curl: (28) Connection timed out after 5000 milliseconds
```

-- FAILED: Confirms public access to backend FastAPI is blocked, which is expected.

e. `curl -v --connect-timeout 5 http://34.47.206.180:9200`
and
`curl -v --connect-timeout 5 http://34.47.206.180:9300`

```
adiborate23@frontend-vm:~/lab08/frontend$ curl -v --connect-timeout 5 http://34.47.206.180:9200
*   Trying 34.47.206.180:9200...
* TCP_NODELAY set
* Connection timed out after 5000 milliseconds
* Closing connection 0
curl: (28) Connection timed out after 5000 milliseconds
adiborate23@frontend-vm:~/lab08/frontend$
```

```
adiborate23@frontend-vm:~/lab08$ curl -v --connect-timeout 5 http://34.47.206.180:9200
*   Trying 34.47.206.180:9200...
* TCP_NODELAY set
* Connection timed out after 5001 milliseconds
* Closing connection 0
curl: (28) Connection timed out after 5001 milliseconds
```

-- FAILED: Confirms public access to Elasticsearch is blocked — again, expected.

f. `curl http://10.160.0.8:9200`
and
`curl http://10.160.0.8:9300`

```
adiborate23@frontend-vm:~/lab08$ curl http://10.160.0.8:9200
curl: (7) Failed to connect to 10.160.0.8 port 9200: Connection refused
adiborate23@frontend-vm:~/lab08$ curl http://10.160.0.8:9300
curl: (7) Failed to connect to 10.160.0.8 port 9300: Connection refused
adiborate23@frontend-vm:~/lab08$
```

-- FAILED: Confirms Elasticsearch cannot be accessed by frontend via private IP too— again, expected because frontend should only communicate with backend (that too only via private IP) and not directly with elasticsearch.

2. Inside backend-vm:

a. `curl https://localhost:9200`

```
adiborate23@backend-vm:~/lab08$ curl http://localhost:9200
{
  "name" : "9043d6c0c80c",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "ebKr7pamTiSJ9ggrwnFtzw",
  "version" : {
    "number" : "7.14.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "dd5a0a2acaa2045ff9624f3729fc8a6f40835aa1",
    "build_date" : "2021-07-29T20:49:32.864135063Z",
    "build_snapshot" : false,
    "lucene_version" : "8.9.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
```

-- WORKED: Confirms Elasticsearch is running and accessible locally from the backend container. Backend can communicate with Elasticsearch.

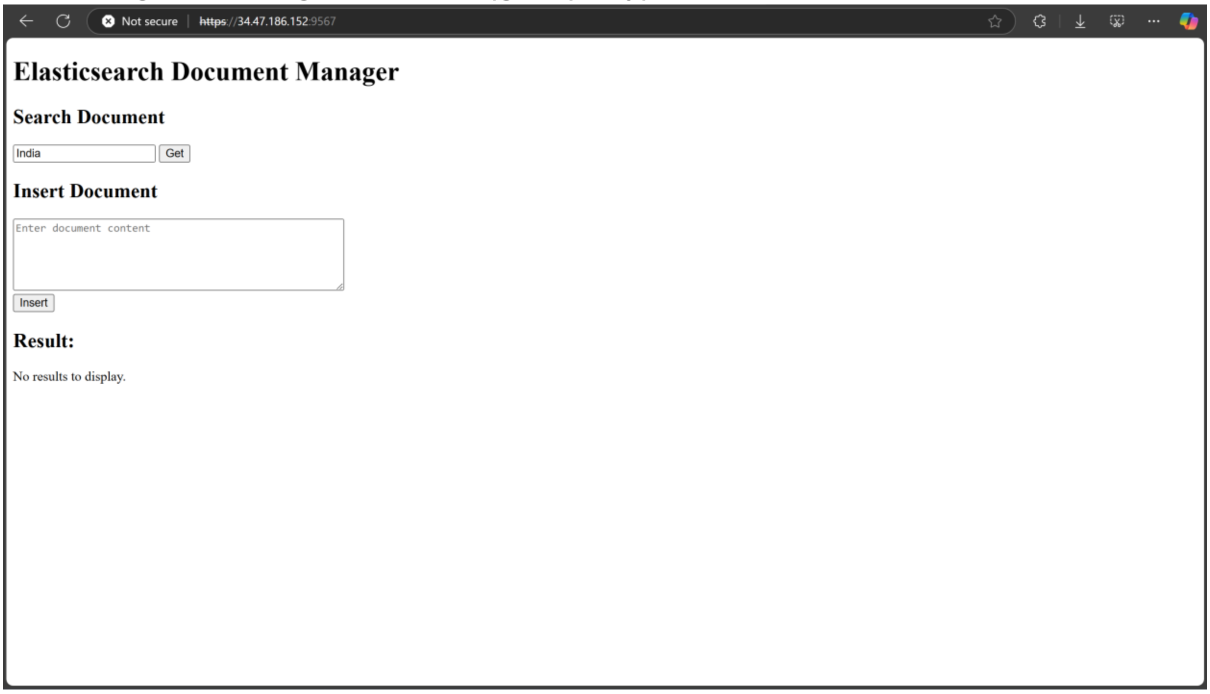
b. `curl -k https://10.160.0.7:9567/docs`

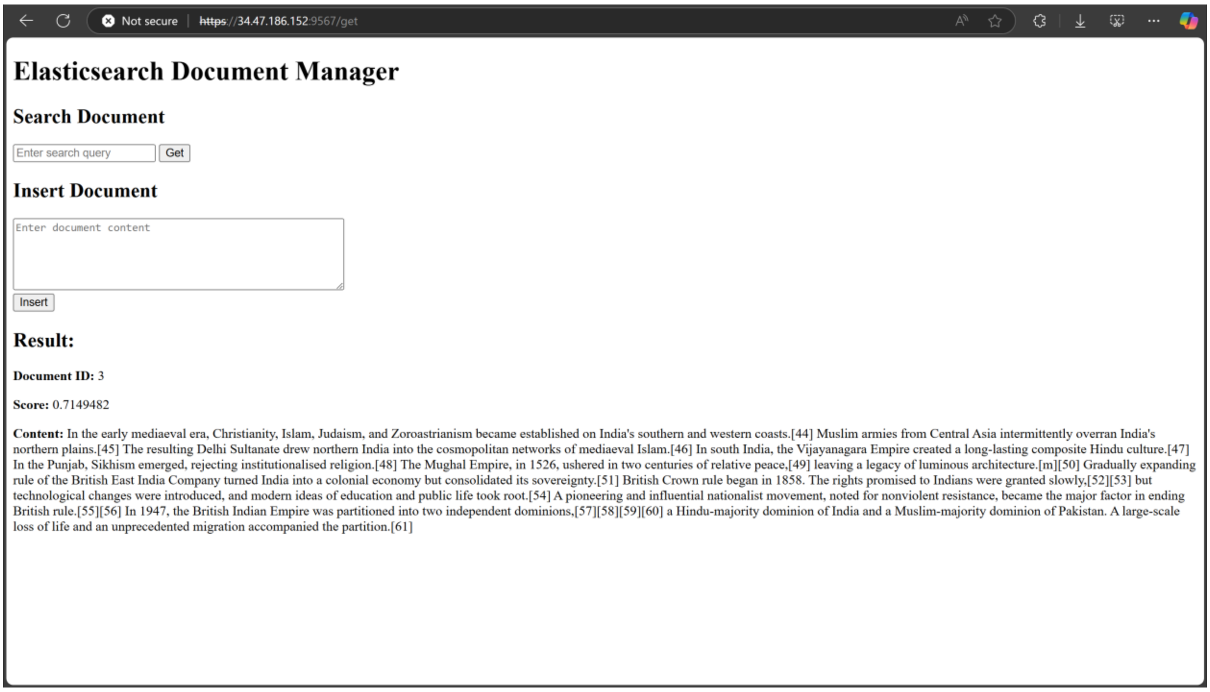
```
adiborate23@backend-vm:~$ curl -k https://10.160.0.7:9567/docs

    <!DOCTYPE html>
    <html>
    <head>
    <link type="text/css" rel="stylesheet" href="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui.css">
    <link rel="shortcut icon" href="https://fastapi.tiangolo.com/img/favicon.png">
    <title>FastAPI - Swagger UI</title>
    </head>
    <body>
    <div id="swagger-ui">
    </div>
    <script src="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui-bundle.js"></script>
    <!-- `SwaggerUIBundle` is now available on the page -->
    <script>
    const ui = SwaggerUIBundle({
        url: '/openapi.json',
    "dom_id": "#swagger-ui",
"layout": "BaseLayout",
"deepLinking": true,
"showExtensions": true,
"showCommonExtensions": true,
oauth2RedirectUrl: window.location.origin + '/docs/oauth2-redirect',
    presets: [
        SwaggerUIBundle.presets.apis,
        SwaggerUIBundle.SwaggerUIStandalonePreset
        ],
    })
    </script>
    </body>
    </html>
    adiborate23@backend-vm:~$
```

-- WORKED: Backend can reach frontend internally via private IP.

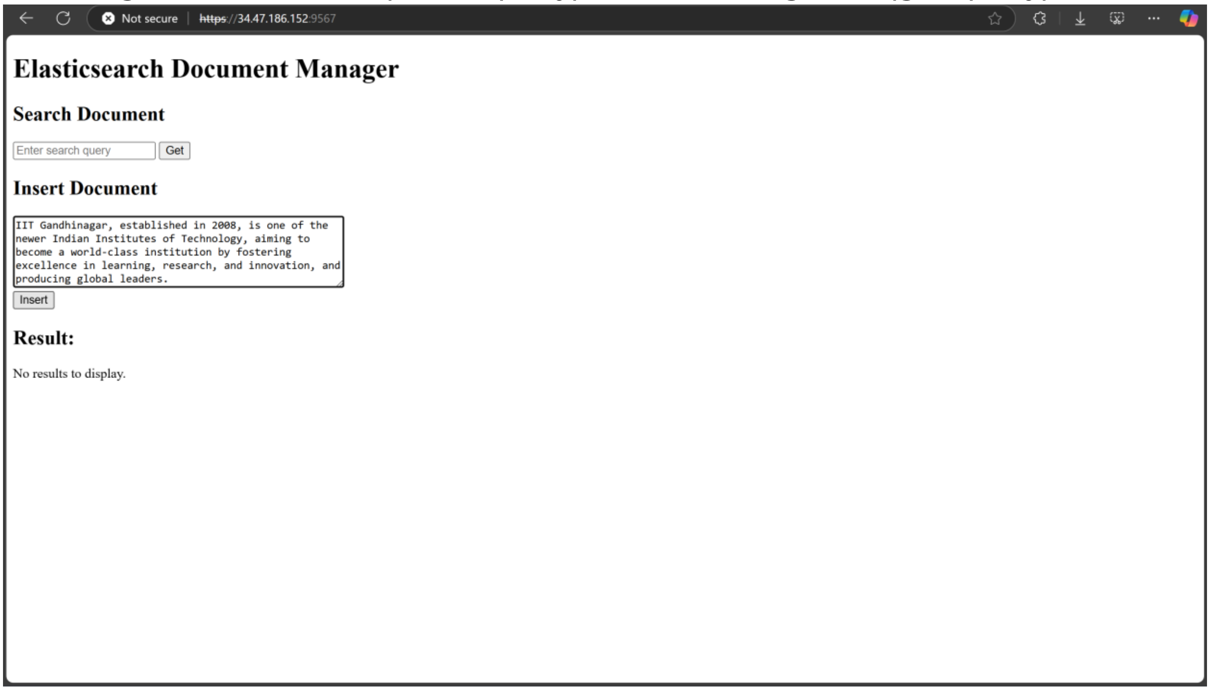
✅ Website Working Demo

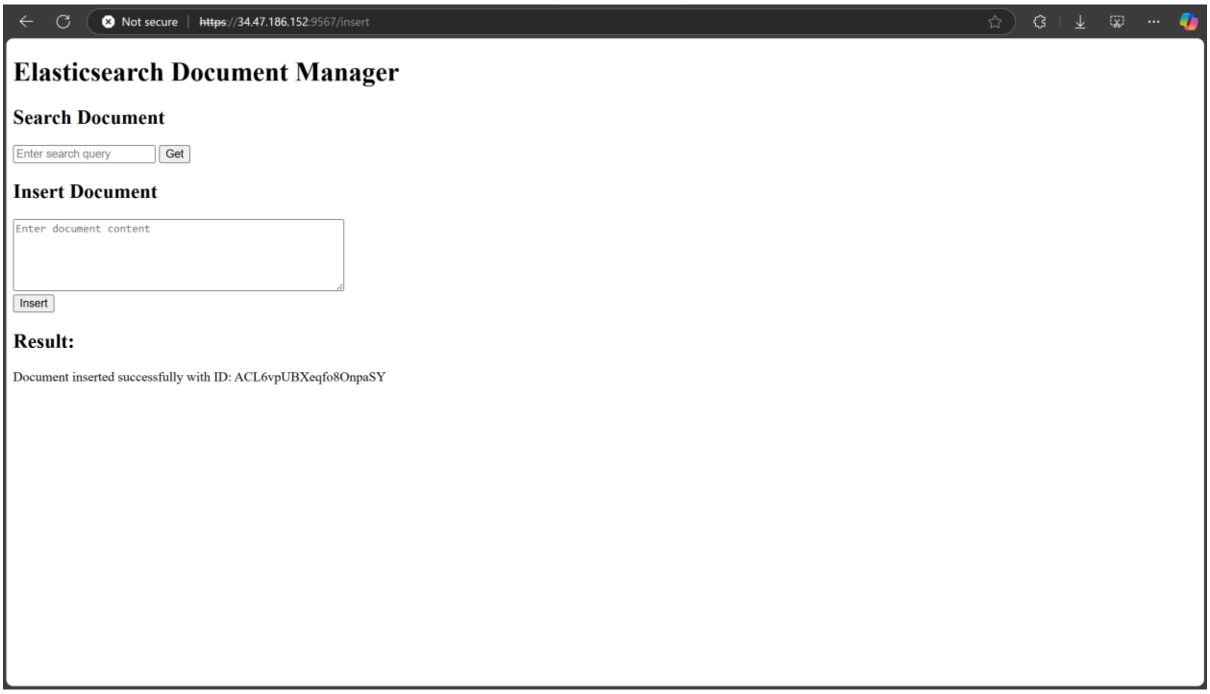
1. Searching in existing document (get query)

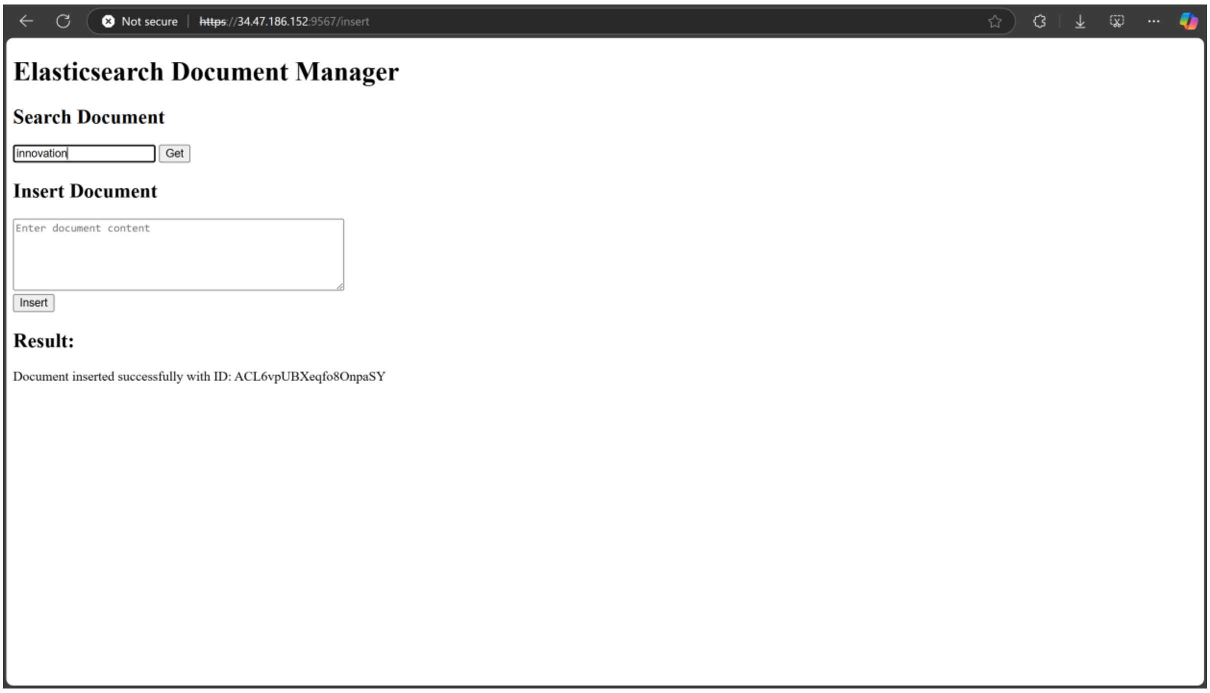


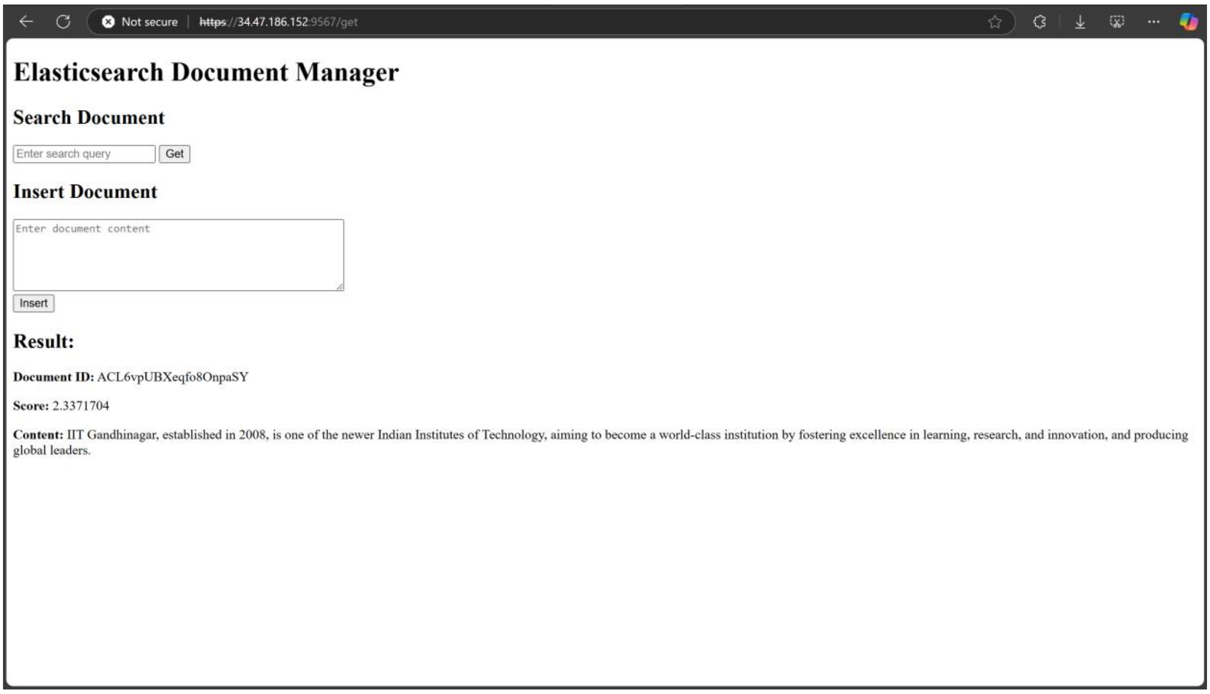


2. Inserting new document (insert query) and searching for it (get query)

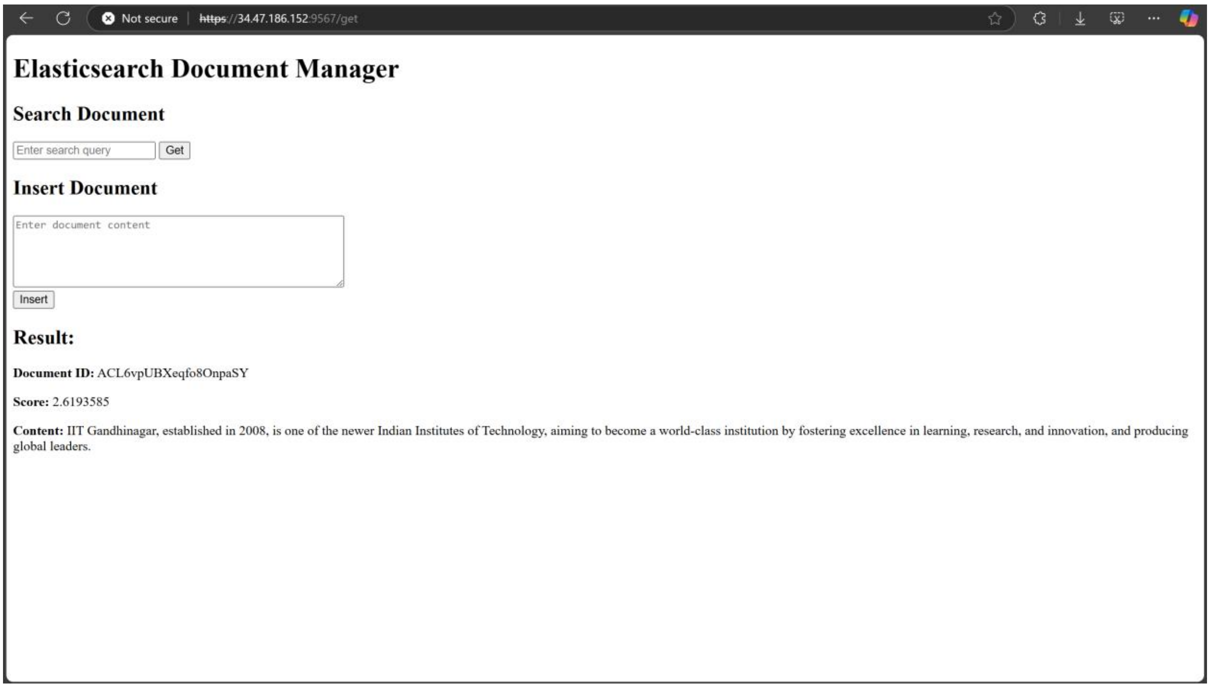



Not secure
https://34.47.186.152:9567/insert
Elasticsearch Document Manager
Search Document
Enter search query
Get
Insert Document
Enter document content
Insert
Result:
Document inserted successfully with ID: ACL6vpUBXeqfo8OnpaSY


Not secure
https://34.47.186.152:9567/insert
Elasticsearch Document Manager
Search Document
innovation
Get
Insert Document
Enter document content
Insert
Result:
Document inserted successfully with ID: ACL6vpUBXeqfo8OnpaSY



3. Checking the ability of elasticsearch to handle typos in queries

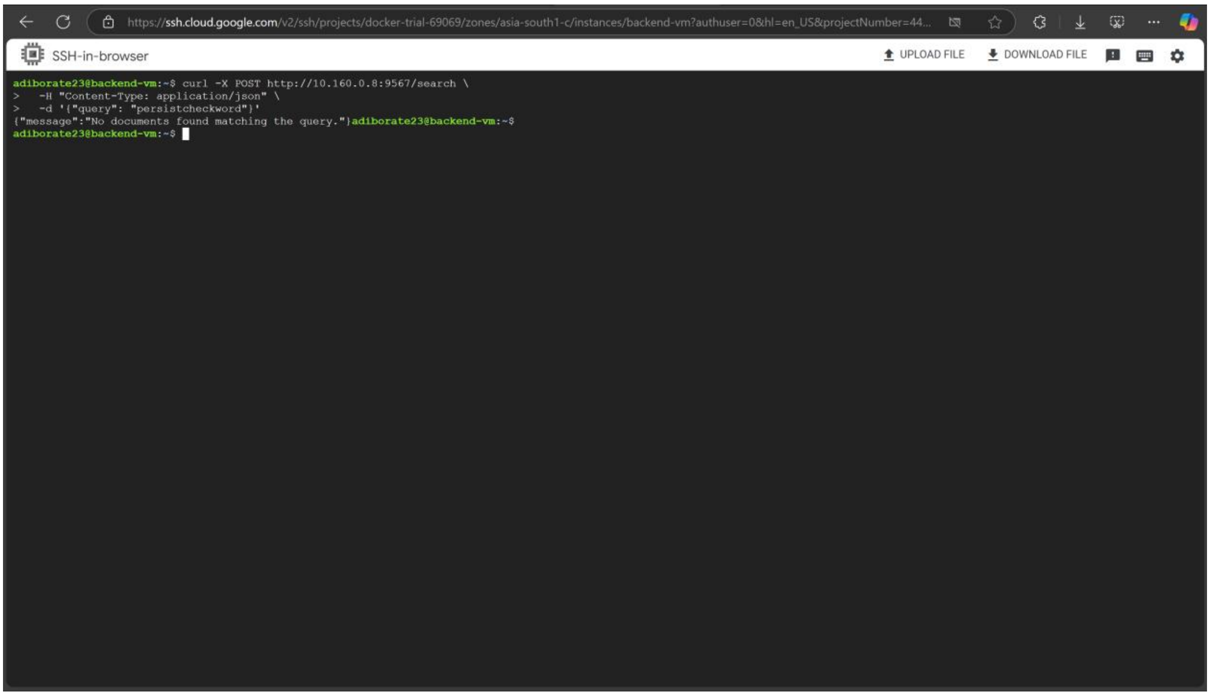


4. Error Handling (Searching for a key that didn't exist in documents)

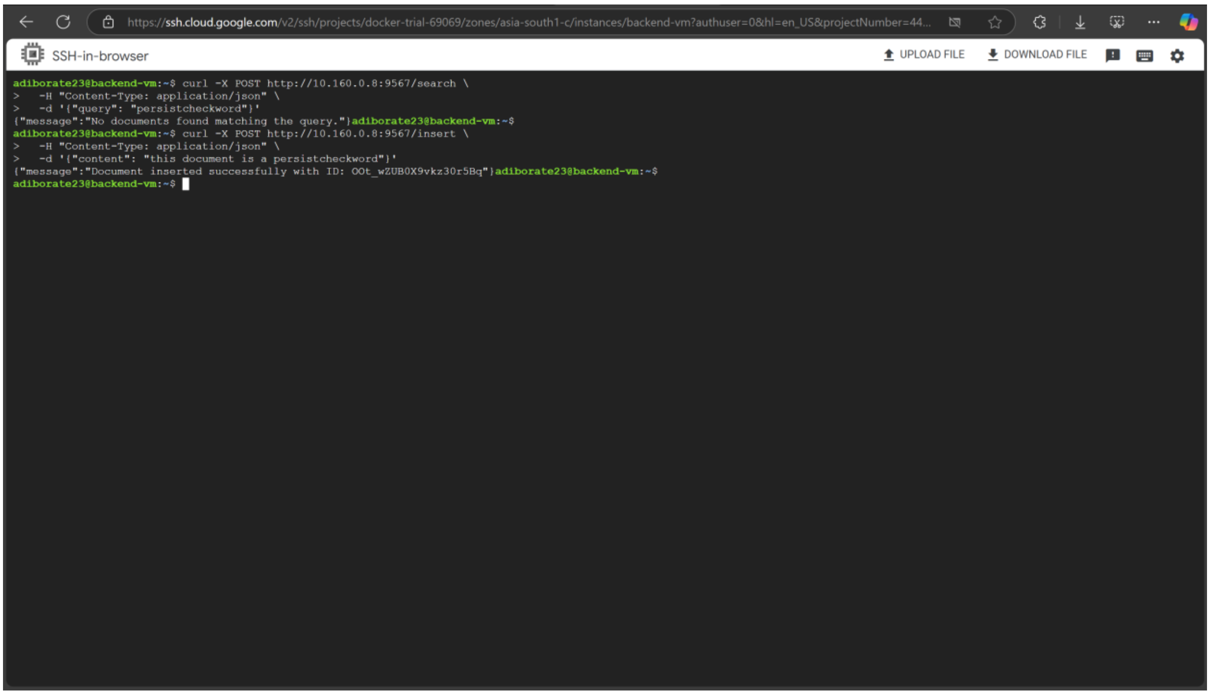
IT WORKED! 👍

✅ Checking data persistence by deleting container

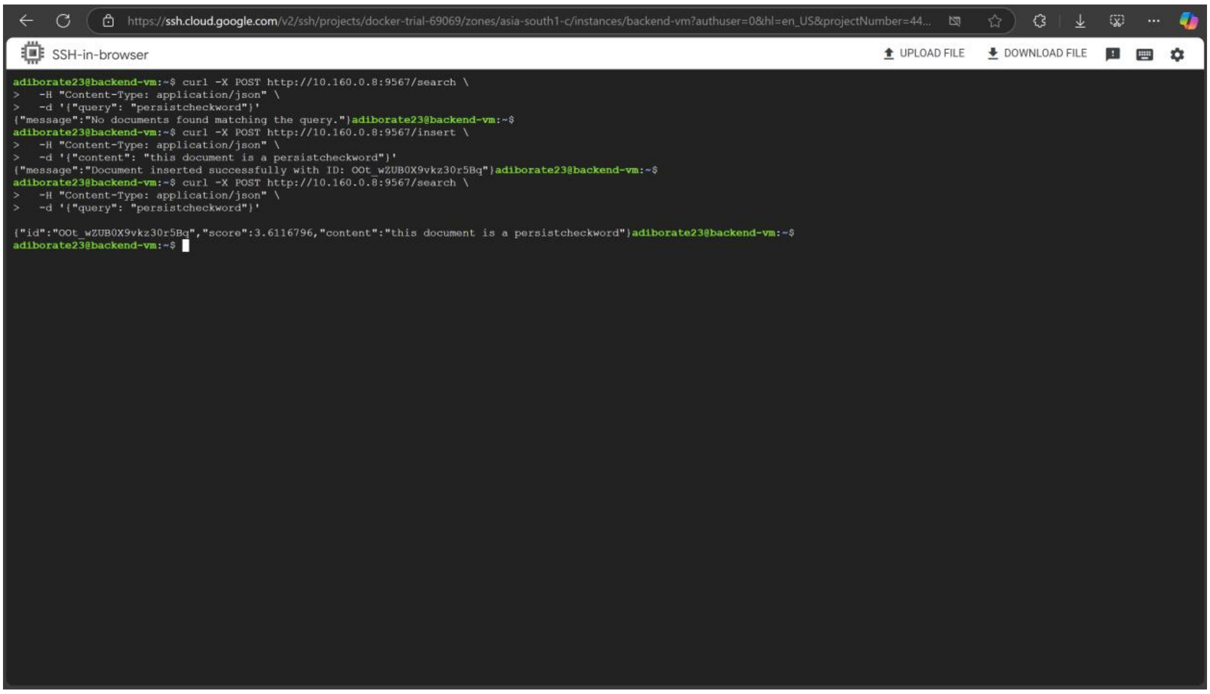
1. Initial Search (Before Insert)



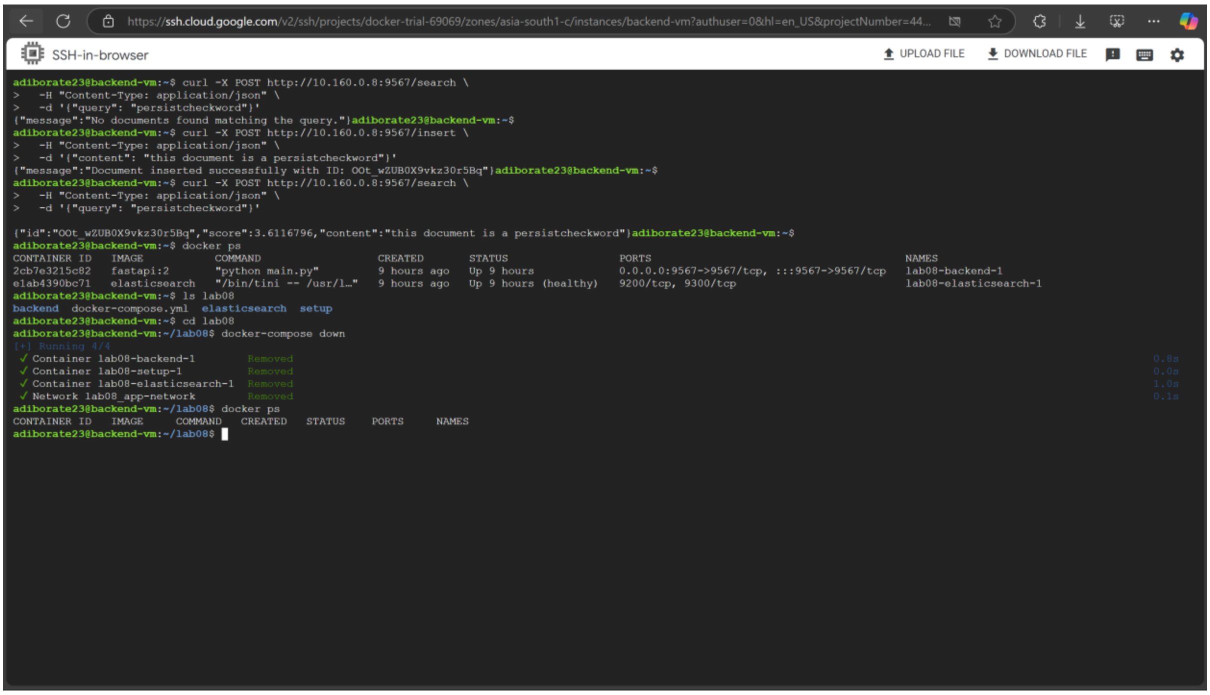
2. Insert a Document

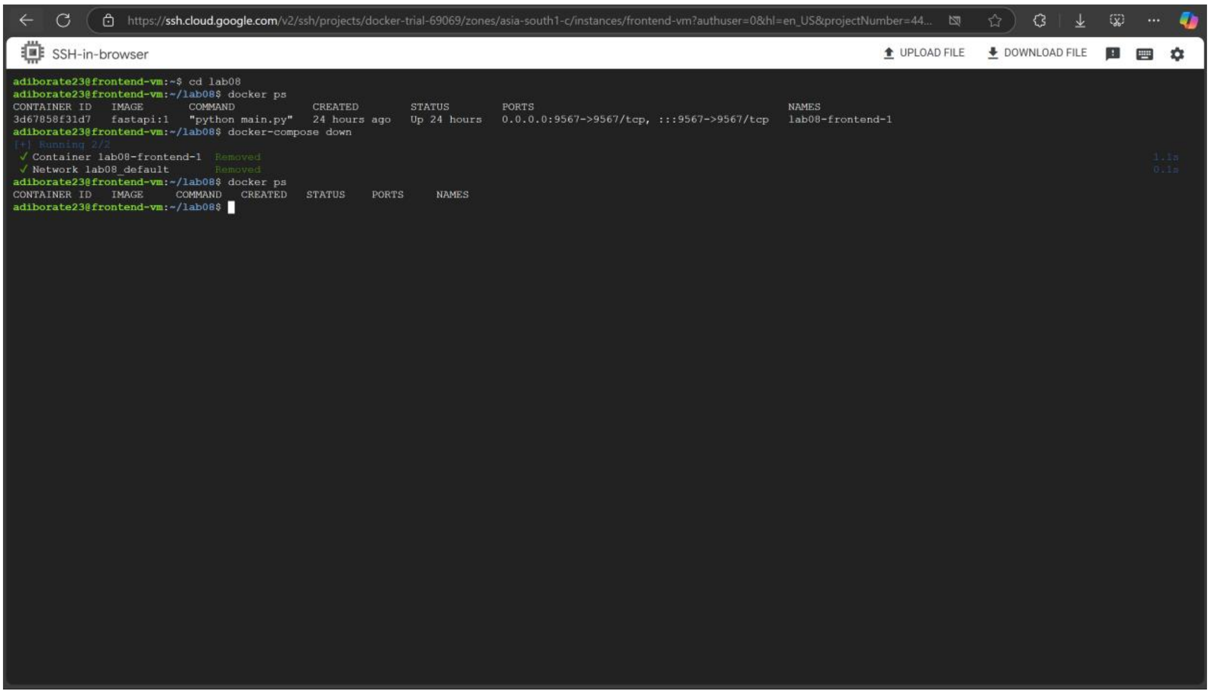


3. Search Again (After Insert)

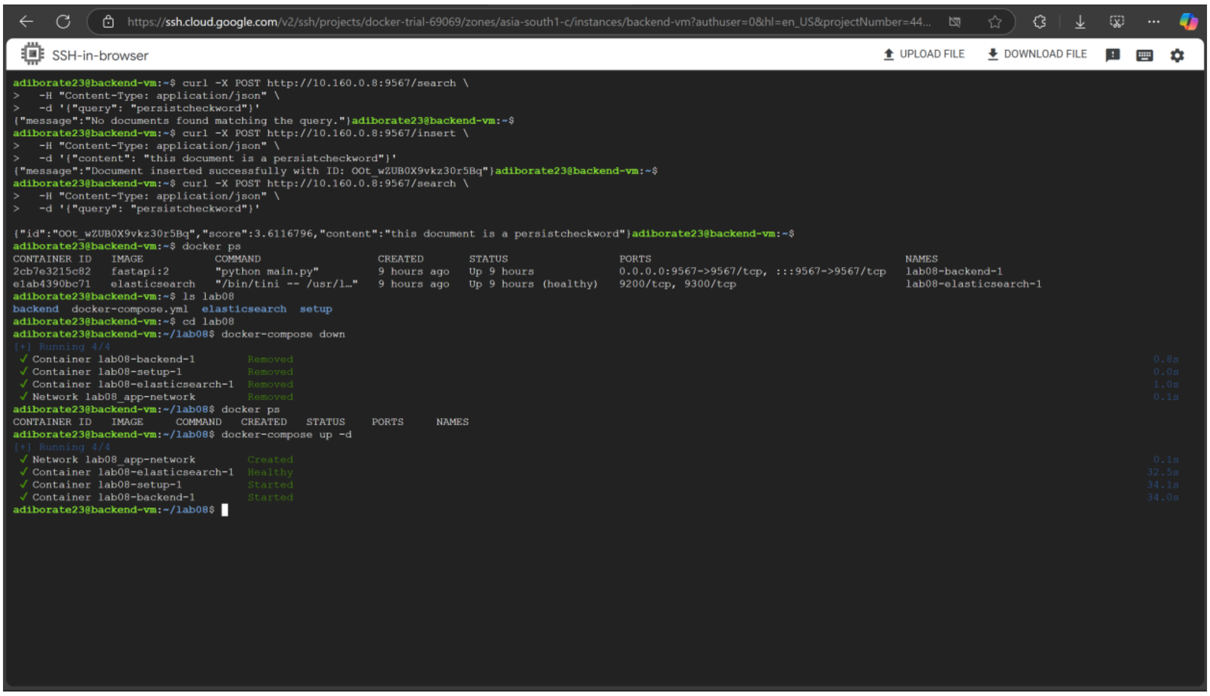


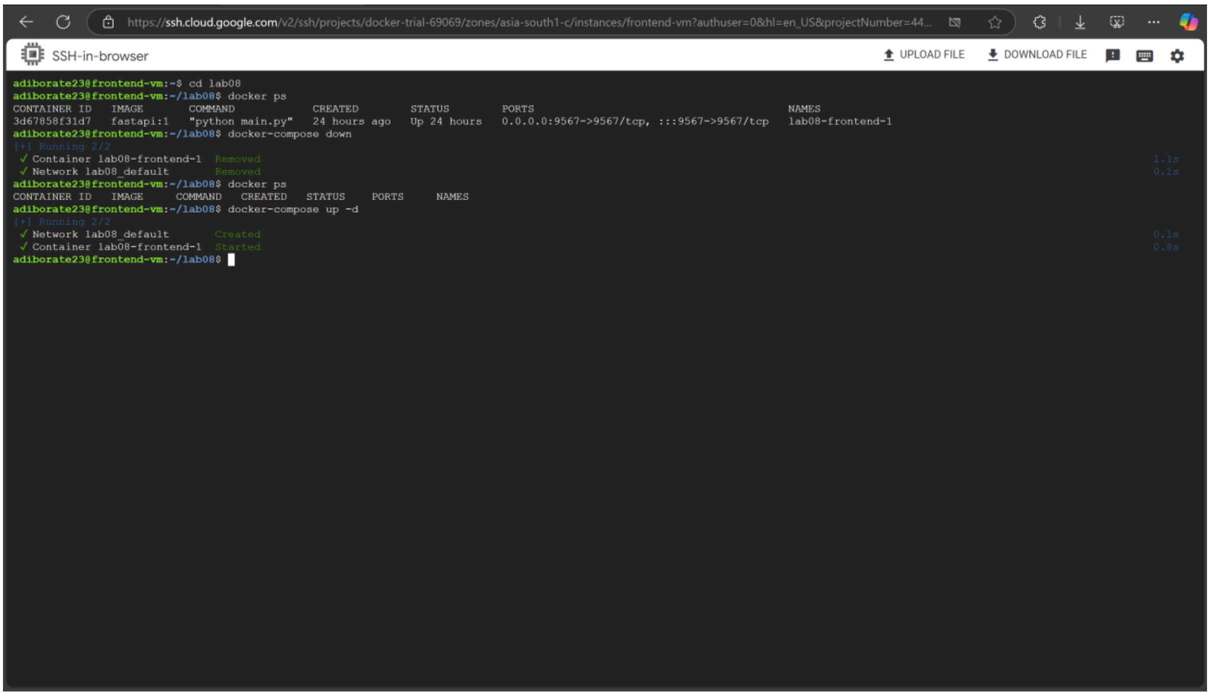
4. Stopping and removing all the containers



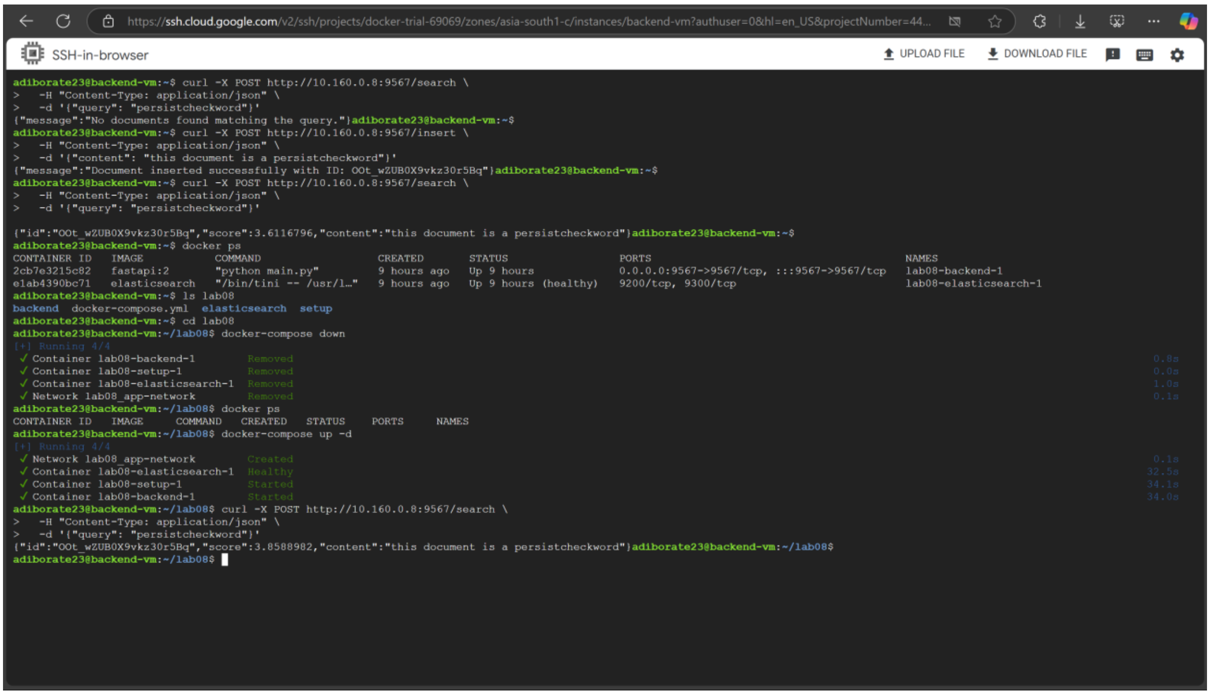


5. Restarting Containers





6. Final Search (After Restarting)

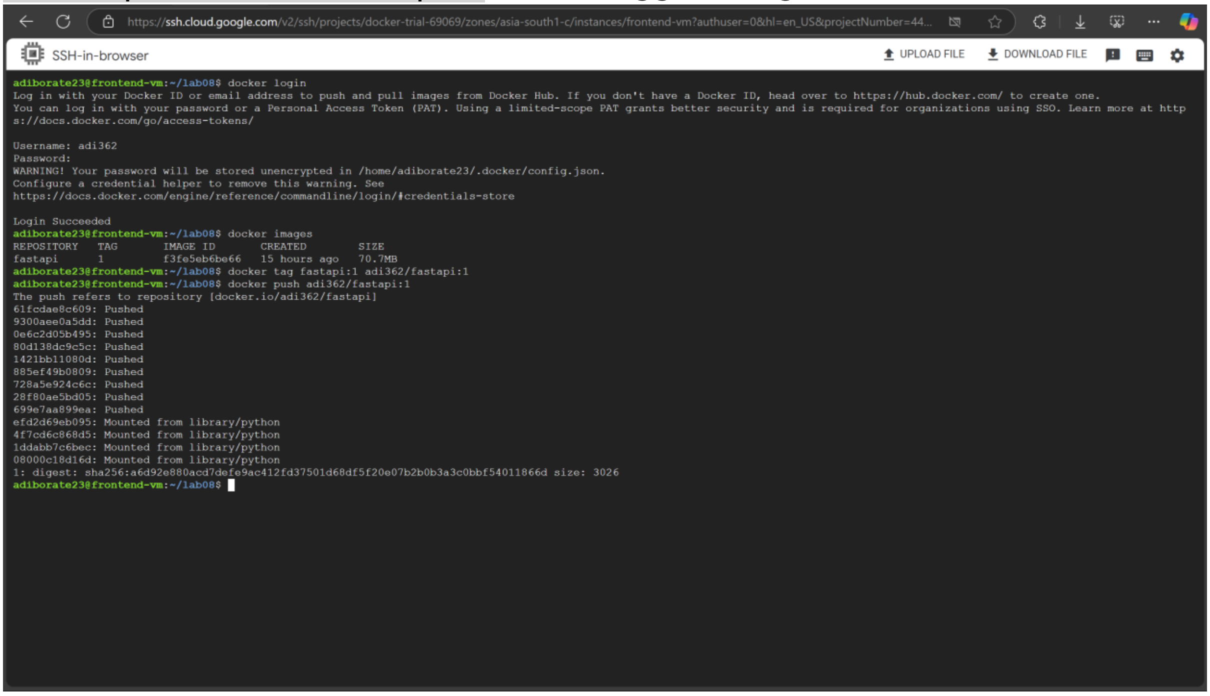


This confirms that data has persisted through container deletion.

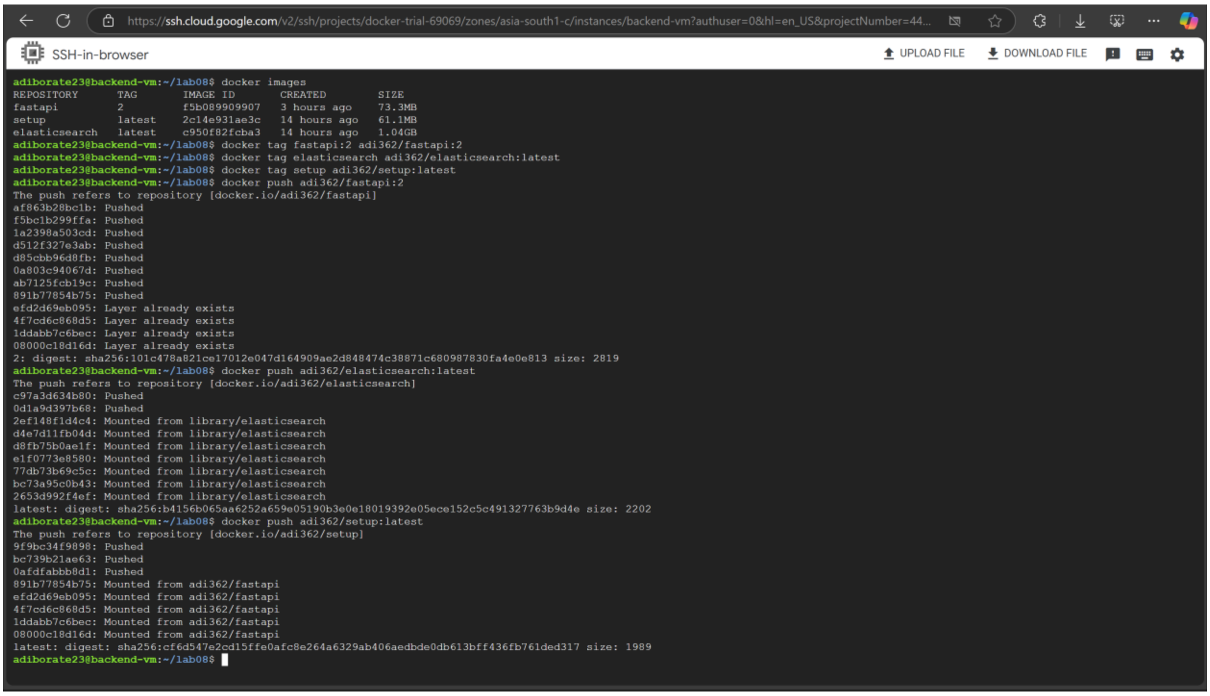
✅ Uploading images created for all three tasks

- From frontend-vm:

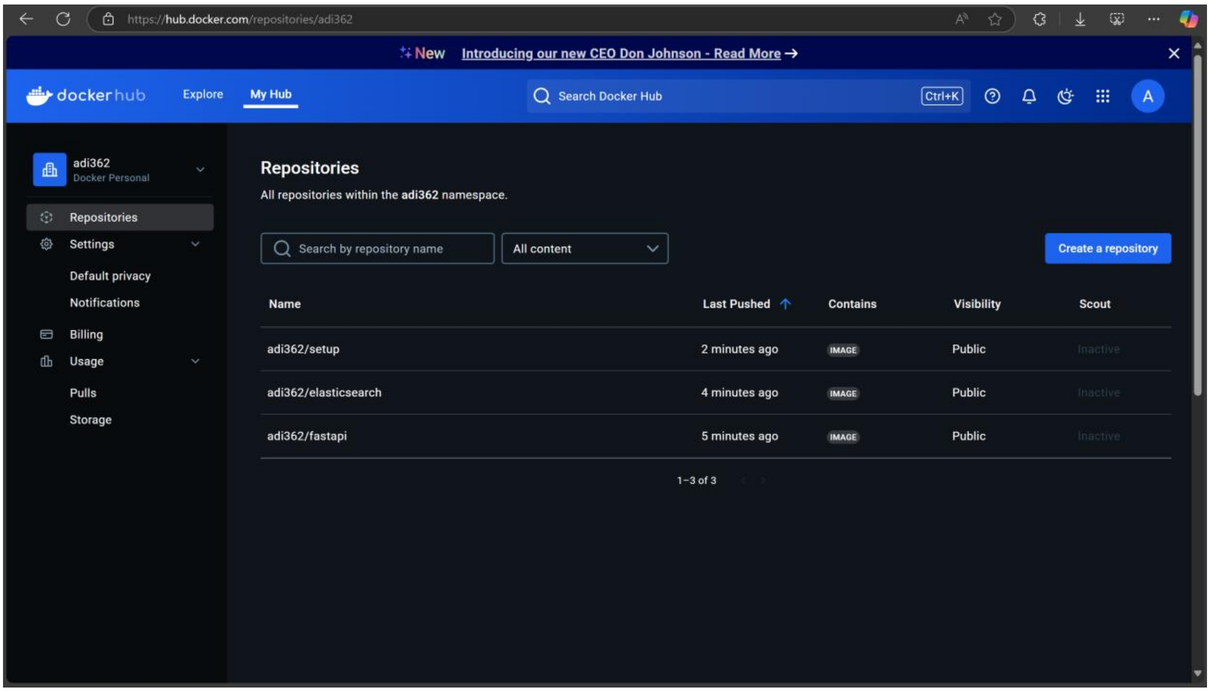
1. `docker login`: to log in into dockerhub account
2. `docker tag fastapi:1 adi362/fastapi:1` : Tag images with my username
3. `docker push adi362/fastapi:1`: Push tagged images to Docker Hub

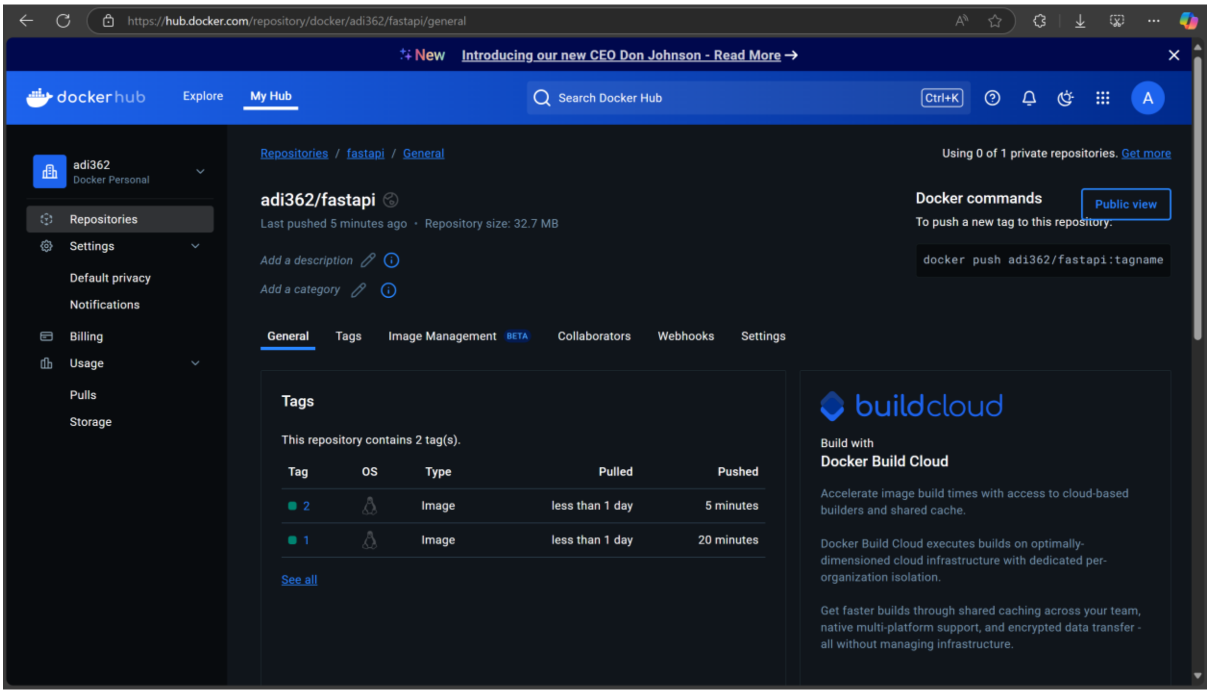


- Similarly from backend-vm:



4. DockerHub Repositories

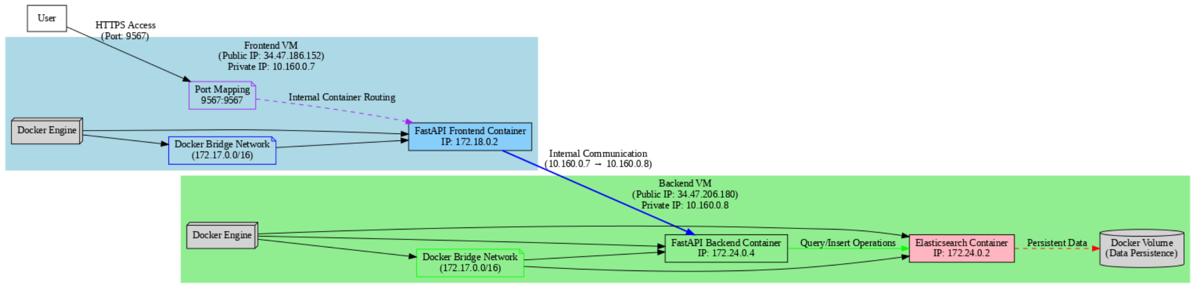




✅ Optimization Checks

- Using python:3.9-alpine — that's already a minimal base image.
- Pip installs are done with --no-cache-dir — avoids unnecessary bloat.
- Non-root user — great security practice.
- Only necessary files are copied — no extra layers or unnecessary bloating.
- No extra tools/libraries or debugging stuff in the final image.
- SSL certs handled properly and permissions adjusted.

✅ Detailed Architecture


User
HTTPS Access
(Port: 9567)
Frontend VM
(Public IP: 34.47.186.152)
Private IP: 10.160.0.7
Port Mapping
9567:9567
Internal Container Routing
Docker Engine
Docker Bridge Network
(172.17.0.0/16)
FastAPI Frontend Container
IP: 172.18.0.2
Internal Communication
(10.160.0.7 → 10.160.0.8)
Backend VM
(Public IP: 34.47.206.180)
Private IP: 10.160.0.8
Docker Engine
Docker Bridge Network
(172.17.0.0/16)
FastAPI Backend Container
IP: 172.24.0.4
Query/Insert Operations
Elasticsearch Container
IP: 172.24.0.2
Persistent Data
Docker Volume
(Data Persistence)